

Email forensic

Module Objectives

- 📁 Email System
- 📁 Email Clients
- 📁 Email Servers
- 📁 Email Message
- 📁 Importance of Electronic Records Management
- 📁 Email Crimes



- 📁 Examples of Email Headers
- 📁 List of Common Headers
- 📁 Why to Investigate Emails
- 📁 Steps to Investigate Email Crimes
- 📁 Email Forensics Tools
- 📁 Laws and Acts against Email Crimes



Module Flow



**Email System
Basics**



Email Crimes



Email Headers



**Steps to
Investigate**



**Email Forensics
Tools**



**Laws and Acts
against Email Crimes**

Email Terminology

IMAP (Internet Message Access Protocol)

It is a method to **access bulletin board messages** or emails residing on a mail server, making them visible and acting as if they were stored locally

BCC (Blind Carbon Copy)

A field that allows you to **send a copy of the message** to a second recipient without notifying the primary recipient

SMTP (Simple Mail Transfer protocol)

It receives outgoing mail from clients and **validates source and destination addresses**, also sends and receives emails to and from other SMTP servers

Attachment

File that is **sent along with an email message** that can be of any type

HTTP (Hypertext Transfer Protocol)

It is used in webmail, and the **message resides** on the webmail server

Email Client

Program used to read and send emails. Typical email clients include **Outlook Express** or **Eudora**

POP3 (Post Office protocol 3)

Standard protocol for receiving email that **deletes mail** on the server as soon as the user downloads it. **Standard port for POP3** is 110

Email Server

A server running at an **Internet Service Provider** or a large website that is connected to the Internet to transport emails. One email server that is typically used is **sendmail** (Mail Transport Agent)

CC (Carbon Copy)

Field in the email header that directs a **copy of the message** to another recipient mail ID

Encoding

Method of **sending binary (non-text) files with emails**. Common encoding options include Uuencode, BinHex, Mime, etc.

Email System

Email system consists of mail clients to **send or fetch emails** and two different SMTP and POP3 or IMAP servers running on a server machine



Email systems are based on a **client-server architecture**

Email is sent from clients to a **central server**, which reroutes the email to its intended destination



Email Clients



- ✂ Email client is a computer application that allows you to send, receive and organize emails
- ✂ Email clients perform the following functions:



Retrieve messages from a mailbox

Allow the user to **create new messages** and submit them to an email server



Display the headers of all the **messages in mailbox**

Allow the user to add **attachments** to the messages they want to send and save the attachments from received messages



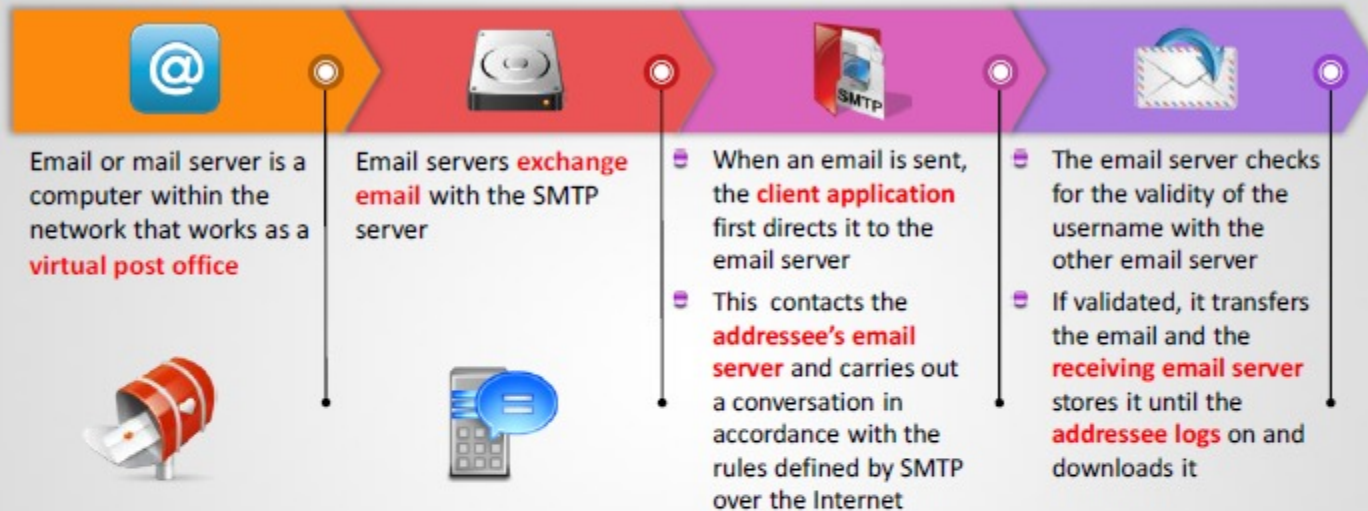
Allow you to select a **message header** and read the body of the email message

Format the messages

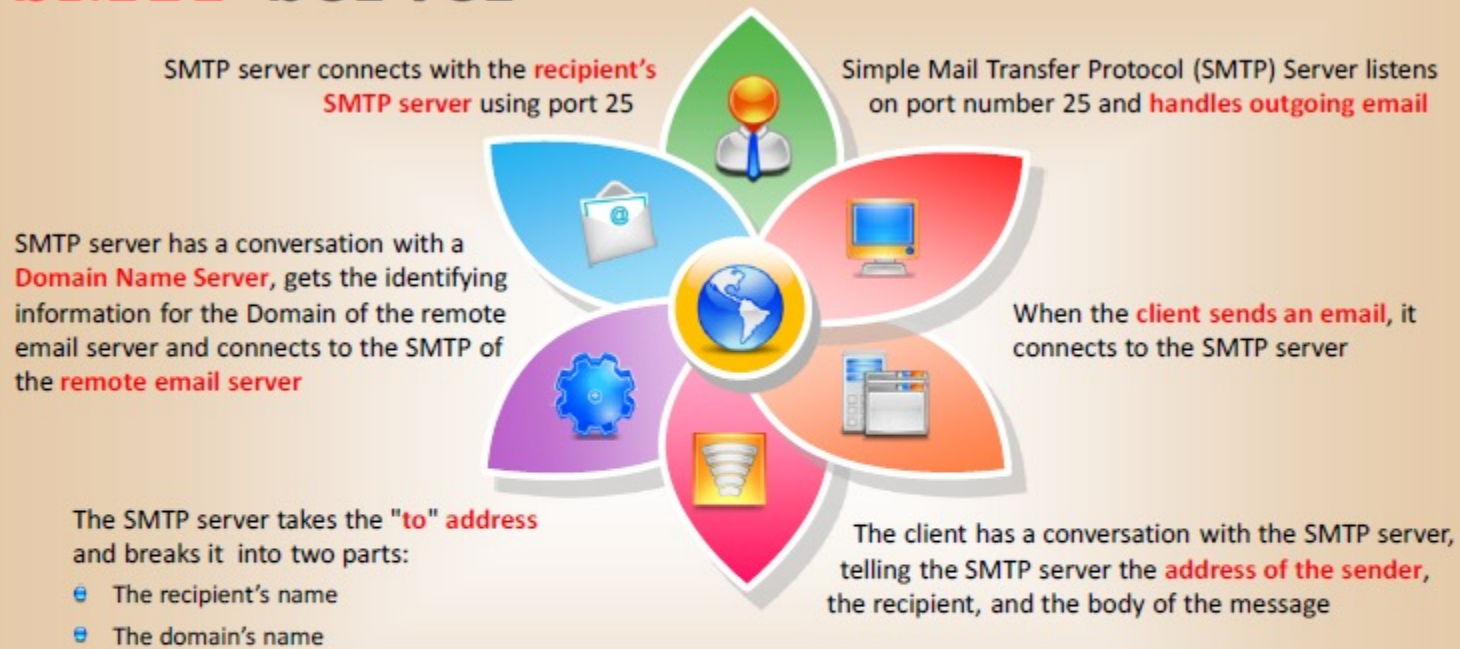
Most commonly used email clients:

- **Microsoft Outlook** and **Thunderbird** (Stand alone)
- **Yahoo! Mail, Gmail, Hotmail, etc.** (Web based)

Email Server



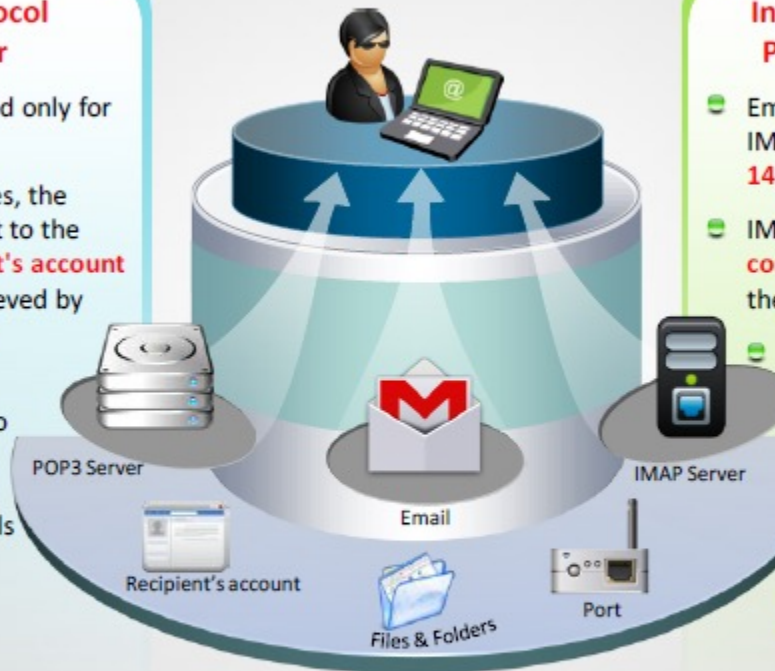
SMTP Server



POP3 and IMAP Servers

Post Office Protocol (POP3) Server

- This email server is used only for incoming emails
- When a message arrives, the POP3 server appends it to the bottom of the **recipient's account file**, which can be retrieved by the email client at any preferred time
- Email client connects to the POP3 server at **port 110** by default to fetch emails



Internet Message Access Protocol (IMAP) Server

- Email client connects to the IMAP server using default **port 143**
- IMAP servers allow **multiple concurrent client connections** to the same mailbox
- It enables accessing and fetching MIME message parts, maintaining message state information at server, creating and manipulating multiple mailboxes on the server and Server-side searches

POP3 can be said to be a **store and forward** service whereas IMAP is a **remote file server**; both are used to read email

Email Message

An email message is composed of three parts:

1. Header

- ✉ Email header contains **information about the email origin** such as the address from where it came, the routing, time of the message, and the subject line
- ✉ Some of the **header information** that is usually important to a technician is kept **hidden** by the email software

2. Body

Body contains the actual message

3. Signature

Used by sender to provide information to the **recipient** about the sender. Email programs can be set to enter this **line automatically** on all the emails sent



Subject: when can we meet?

Date: Mon, 6, Aug 2011 10:04:22 -0500

From: alvernja@alverno.edu

To: anthosb@alverno.edu

Header

When can we get together to work on our project? I am available any time this week after 5:00 PM. But I do have some other appointments next week. I would like to meet before we have our next class so email me and let me know what would work for you.

Body

Thanks!

Jane A. Alverno
Student, Alverno College

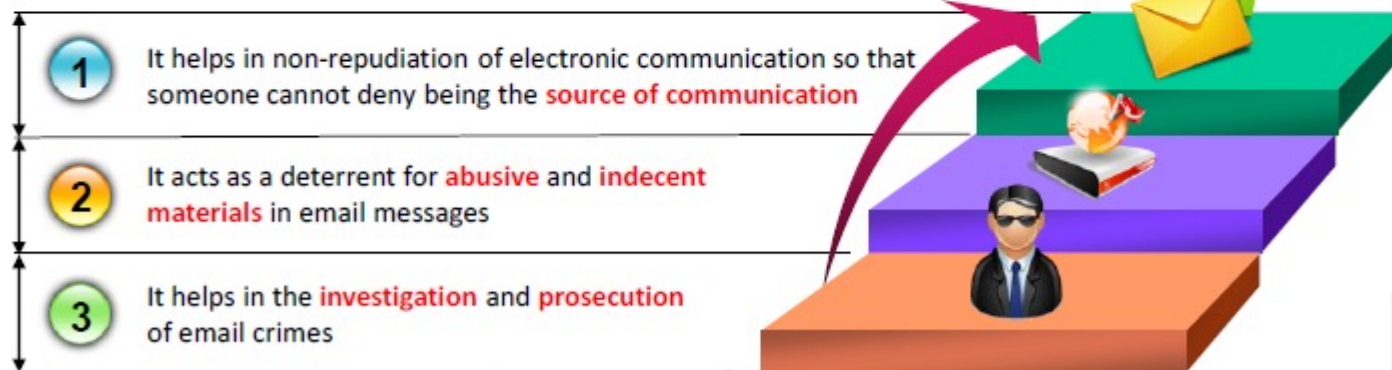
Signature

Importance of Electronic Records Management

Electronic records management may be defined as The field of management responsible for the **efficient** and **systematic control** of the creation, receipt, maintenance, use and disposition of electronic records, including the **processes** for **capturing** and **maintaining evidence** of and information for legal, fiscal, administrative, and other business purposes



Importance of electronic records management:



Module **Flow**



**Email System
Basics**



Email Crimes



Email Headers



**Steps to
Investigate**



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**Laws and Acts
against Email Crimes**

Email Crime

- Ease, speed, and relative anonymity of email makes it a potent tool for criminals



Email Spamming

- Spamming can be defined as **sending unsolicited emails**
- Spammers obtain **email addresses** by gathering addresses from Usenet postings, DNS listings, or web pages
- User with a **valid email address** can spam email address, bulletin-board services, or newsgroups

Common Subject
headers of Spam
mails



Search:		Status:	Any Status
Subject	Sender	Date	
check this out man...	Nelda Romano	Thursday 14:59:37	
Help me!	Osvaldo MANNING	Thursday 12:47:59	
Have Arthritis pains? There is help for you.	Orsa	Thursday 03:45:36	
down on her, and	Reginald Stubbs	Wednesday 06:02:05	
natural enlargement	diane george	Tuesday 16:37:15	
No Subject	fabian dickhaut	Monday 10:38:59	
only Youngest have Shocking	Kristie Sapp	Monday 01:07:32	
Reduces stress	frankie kim	06.02.2005 16:27	
PERSONAL	esno2005	06.02.2005 04:56	
We need to render the delight of having the finest	Clotilda Gadnugt	06.02.2005 02:10	
Find more savings online	kennith draper	05.02.2005 22:30	
faster cheaper meds	Lidia White	05.02.2005 16:37	
Breaking News	Dee H. Edwards d	05.02.2005 14:40	
We have your wanted meds at low prices only.	lucien hyatt	04.02.2005 06:59	
100% zum einladen__1679438	Isel Rios	03.02.2005 03:34	
Enjoy your wanted meds.	tracey uliano	03.02.2005 02:28	
Confirm Your Washington Mutual Online Banking	Washington Mutual On...	02.02.2005 22:03	
out PINNACLE SYSTEM, MACROMEDIA, SYMANTEEC, PC GAMES, ...	Valerie Ileen	02.02.2005 19:11	
Finished	Cecilia Fuller	02.02.2005 06:57	
You can save more thru ordering meds on our site.	mel sevic	02.02.2005 01:21	

Mail Bombing/Mail Storm



Mail Bombing

- ☛ Sending huge volumes of email to an address in an attempt to **overflow the mailbox** or **overwhelm the server** where the email address is hosted to cause a **denial-of-service attack**
- ☛ In many instances, the messages will be large and constructed from meaningless data in an effort to consume additional system and network resources



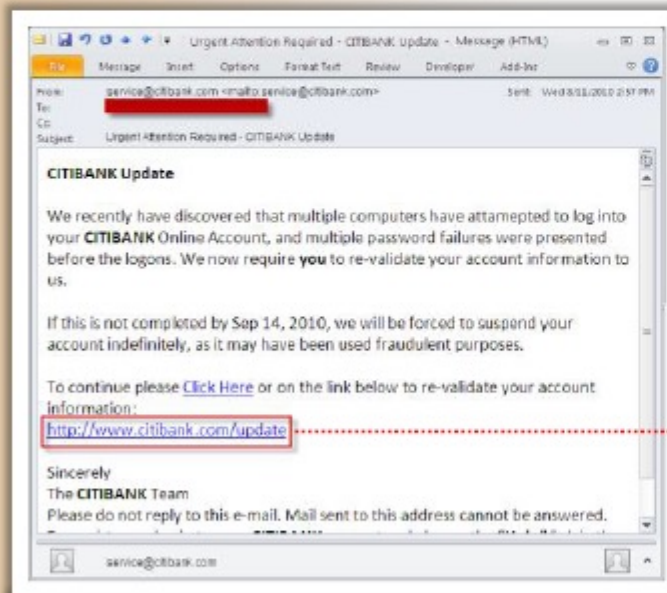
Mail Storm

- ☛ It is a sudden spike of **"Reply All"** messages on an email distribution list, caused by **one misdirected message**

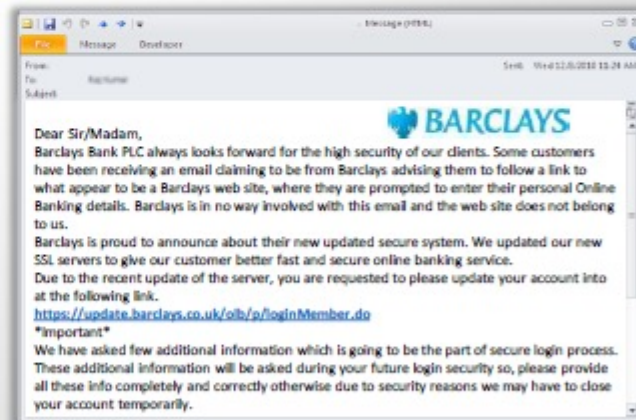
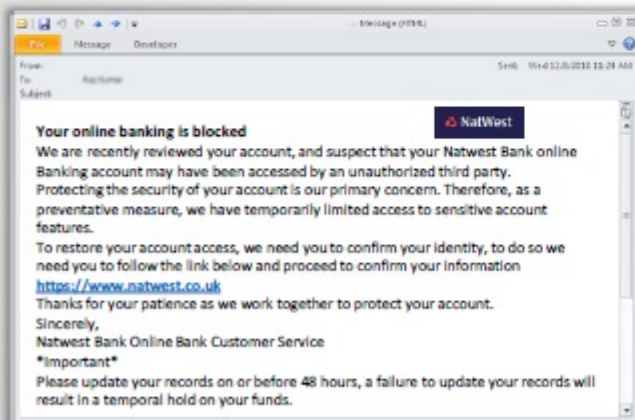
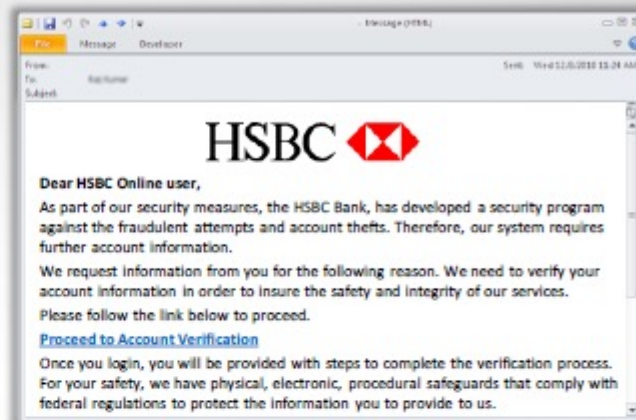
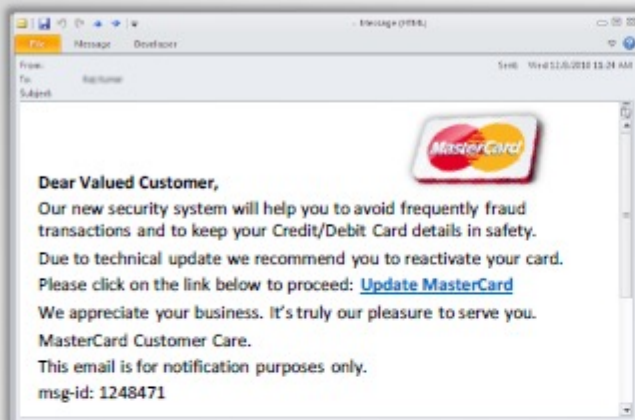


Phishing (Cont'd)

- It is a criminal act of sending an **illegitimate email**, falsely claiming to be from a legitimate site in an attempt to acquire the user's personal or account information
- Phishing emails redirect users to **fake webpages** of trustworthy sites that ask them to submit their personal information
- Phishing attacks can target millions of email addresses around the world using **mass-mailing systems**



Phishing

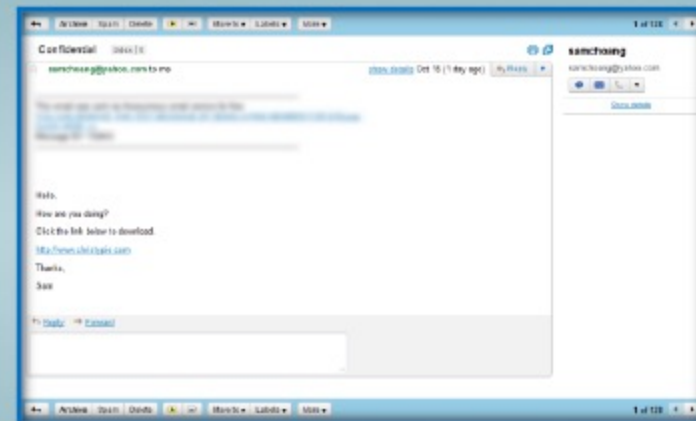


Email Spoofing

- ❌ Email spoofing is the **forgery of an email header** so that the message appears to have originated from someone or somewhere other than the actual source
- ❌ **Spammers and perpetrators of phishing change the email header fields** such as the From, Return-Path, and Reply-To-Fields to hide the actual source

Example: Using the Anonymailer service, Mr. Smith, whose email address is smith@hotmail.com, can send viruses, Trojans, worms, etc. to Sam's friends using Sam's email address samchoang@yahoo.com. Sam's friends would then download them believing them to be from a trustworthy source

The screenshot shows a web form titled "Send Fake Anonymous Email". It has fields for "From Name" (empty), "From E-mail" (containing "samchoang@yahoo.com" and highlighted with a red box), "To" (containing "smith@hotmail.com"), and "Subject" (containing "Confidential"). Below these is a rich text editor with the following content: "Hello, How are you doing? Click the link below to download: http://www.schuyler.com Thanks, Sam". At the bottom, there is a CAPTCHA section with the text "Are you human? Type the characters on the image to the form field below." and an image showing the characters "GH15LK". A "Submit" button is at the bottom right, with a note "(*) Mandatory fields".



A chat room is a website, part of a website, or part of an online service that provides a **venue for communities** of users with a common interest to communicate in real time



They can also be used as a **social engineering tool** to collect information for committing several other crimes

Crime via Chat Room

Chat rooms are **increasingly being used for a variety of crimes** such as child pornography, cyber stalking, and identity thefts

They are a regular feature of different adult sites and are extensively used to **disseminate obscene materials** over Internet

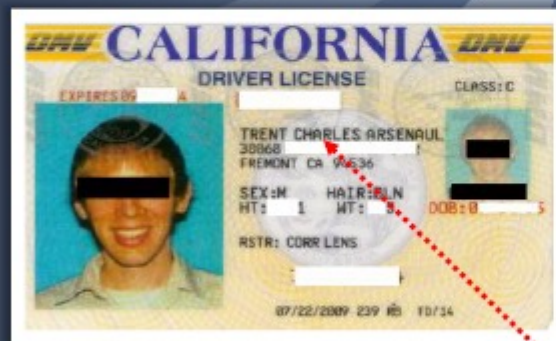
Identity Fraud/Chain Letter

Identity Fraud

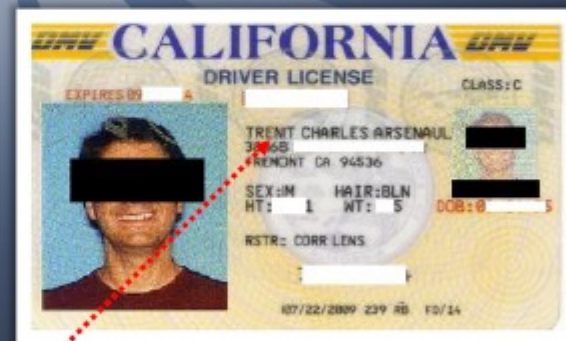
Identity fraud is used to refer to all types of crime in which someone **wrongfully obtains and uses another person's personal data** in a way that involves fraud or deception for economic gain

Chain Letter

A chain letter by definition is a letter **directing the recipient to send out multiple copies** so that its circulation increases exponentially



Original



Identity Theft

Same Name: TRENT CHARLES ARSENAUL

Module Flow



**Email System
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Email Crimes



Email Headers



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Example of Email Header

Header Line	Meaning
Microsoft Mail Internet Headers Version 2.0	Email header title
Received: from EXIC1.lse.ac.uk ([132.148.290.111]) by ExF2.lse.ac.uk with Microsoft SMTPSVC(5.0.2195.5329); Tue, 2 Nov 2010 12:20:40 +0100	Email passed from Exchange gateway servers to staff mailbox Server
Received: from EXAV2.pc.lse.ac.uk ([132.148.290.123]) by EXIC1.lse.ac.uk with Microsoft SMTPSVC(5.0.2195.5329); Tue, 2 Nov 2010 12:20:39 +0100	Email passed from anti-virus servers to Exchange gateway Server
Received: From exas1.lse.ac.uk ([132.148.290.127]) by EXAV2.pc.lse.ac.uk (WebShield SMTP v4.5 MR1a); id 1058267813844; Tue, 2 Nov 2010 12:20:37 +0100 Content-Class: urn:content-classes:message X-MimeOLE: Produced By Microsoft MimeOLE V5.50.4910.0300	Email passed from anti-spam servers to anti-virus servers
Received: from web60003.mail.yahoo.com ([120.204.138.115]) by exas1.lse.ac.uk with Microsoft SMTPSVC(5.0.2195.5329); Tue, 2 Nov 2010 12:18:20 +0100	Email received by LSE anti-spam server from originator's email server
Message-ID: <2003056893256.6388.qmail@web60003.mail.yahoo.com>	
Received: from [132.148.290.30] by web60003.mail.yahoo.com via HTTP; Tue, 2 Nov 2010 12:18:20 BST	Date and time received by the originator's email servers
Date: Tue, 2 Nov 2010 12:18:20 +0100 (BST)	Date and time sent
From: "Daniel Simpson" <djs@yahoo.co.uk>	Originator's email address
Reply-To: <d.simpson@hotmail.com>	Reply to address
Subject: Test Email	Subject
To: <f.muir@lse.ac.uk>	Recipients email address
MIME-Version: 1.0 Content-Type: multipart/alternative; boundary="0-714528637-1235867664=:1234" Content-Transfer-Encoding: 8bit	
Return-Path: <djs@yahoo.co.uk> X-OriginalArrivalTime: 2 Nov 2010 11:20:20.0570 (UTC) FILETIME=[3F7D6840:D1C34AC2]-00-714528637-1235867664=:1234 Content-Type: text/plain; charset="iso-8859-1" Content-Transfer-Encoding: 7bit-0-714528637-1235867664=:1234 Content-Type: text/html; charset="iso-8859-1" Content-Transfer-Encoding: 7bit-0-714528637-1235867664=:1234--	Originator's email address Arrival time at originators email servers Start of the email message

List of Common Headers (Cont'd)

Apparently-To

- Messages with many recipients sometimes have a long list of headers in the form "**Apparently-To:** **rth@bieberdorf.edu**" (one line per recipient)
- These headers are **unusual in legitimate emails**; they are normally a sign of a mailing list, and in recent times mailing lists have generally used software not sophisticated enough to generate a giant pile of headers



Bcc

- BCC stands for "**Blind Carbon Copy**". It is used to add addresses to the SMTP delivery list but not (usually) listed in the message data, remaining invisible to other recipients.
- The idea is to be able to send **copies of email** to persons who might not want to receive replies or to appear in the headers
- Blind carbon copies are popular with **spammers**, since it confuses many inexperienced users to get email that does not appear to be addressed to them





List of **Common Headers** (Cont'd)

Cc

- Cc stands for "**Carbon Copy**"
- This header is sort of an extension of "**To:**"; it specifies **additional recipients**. The difference between "**To:**" and "**Cc:**" is essentially connotative; some email programs also deal with them differently in generating replies



Comments

- This is a nonstandard **header** field. Example: "Comments: Authenticated sender is <**rth@bieberdorf.edu**>"
- It is added by some mailers to **identify the sender**; however, it is often added by hand by spammers as well

List of **Common Headers** (Cont'd)

1

Content-Transfer-Encoding: This header relates to MIME, a **standard way of enclosing non-text content** in email; it has no direct relevance to the delivery of email, but it affects how MIME-compliant email programs interpret the content of the message



2

Content-Type: Another MIME header, telling **MIME-compliant email programs** what type of content to expect in the message



3

Date: This header does exactly what you expect; it specifies a date, normally the date the message was composed and sent. If this header is **omitted by the sender's computer**, it might conceivably be added by an email server or even by some other machines along the route



4

Errors-To: Specifies an address for **mailer-generated errors**, like "no such user" bounce messages, to go to (instead of the sender's address). This is not a particularly common header, as the sender usually wants to receive any errors at the sending address, which is what most (essentially all) **email server software** does by default



List of **Common Headers** (Cont'd)

From: (with colon) Denotes the sender's email address



In-Reply-To: A Usenet header that occasionally appears in email, the **In-Reply-To:** header gives the message ID of some previous message which is being replied to. It is unusual for this header to appear except in email directly related to Usenet; spammers have been known to use it, probably in an attempt to evade filtration programs



Message-Id: It is a unique identifier assigned to each message, by the first email server it encounters. Conventionally, it is of the form "**gibberish@bieberdorf.edu**", where the "gibberish" part could be absolutely anything and the second part is the name of the machine that assigned the ID. Sometimes, the "gibberish" includes the sender's username. Any email in which the message ID is malformed or in which the site in the message ID is not the real site of origin, is probably a forgery



Mime-Version: (also MIME-Version:) Yet another MIME header, this one just specifying the version of the **MIME protocol** that was used by the sender. Like the other MIME headers, this one is usually ignorable; most modern email programs will do the right thing with it



List of Common Headers (Cont'd)

Newsgroup



- This header only appears in **email** that is connected with **Usenet**—either email copies of Usenet postings, or email replies to postings
- In the first case, it specifies the newsgroup(s) to which the **message was posted**; in the second, it specifies the newsgroup(s) in which the **message being replied to was posted**
- The semantics of this header are the subject of a low-intensity holy war, which effectively assures that both **sets of semantics** will be used indiscriminately for the foreseeable future

Reply-To



- Specifies an address for replies to go to. Though this header has many **legitimate uses** (perhaps your **software mangles** your From: address and you want replies to go to a correct address), it is also widely used by spammers to **deflect criticism**
- Occasionally, a naive spammer will actually solicit responses by email and use the Reply-To: header to collect them, but more often the Reply-To: address in **junk email** is either invalid or an innocent victim

References



- The References header is rare in email except for copies of **Usenet postings**
- Its use on Usenet is to identify the **"upstream"** posts to which a message is a response; when it appears in email, it is usually just a copy of a Usenet header
- It may also appear in **email responses** to Usenet postings, giving the **message ID** of the post being responded to as well as the references from that post

List of **Common Headers** (Cont'd)

Priority

- It is a free-form header that assigns a priority to the email, but most software ignore it
- It is often used by **spammers**, usually in the form "**Priority: urgent**" (or something similar), in an attempt to get their messages read



Organization

- It is a completely free-form header that normally contains the name of the organization through which the sender of the message has net access
- The **sender** can generally **control this header**, and **silly entries** like "Royal Society for Putting Things on Top of Other Things" are commonplace



Sender

- This header is unusual in email (X-Sender: is usually used instead), but appears occasionally, especially in **copies of Usenet posts**
- It should identify the sender; in the case of Usenet posts as it is a more **reliable identifier** than the From: line



List of Common Headers (Cont'd)



Subject:

- A completely **free-form field** specified by the sender, intended, of course, to describe the **subject of the message**

X-Confirm-Reading-To:

- This header requests an **automated confirmation** notice when the message is received or read
- It is typically ignored; presumably some software acts on it



X-Mailer (also X-mailer):

- This is a freeform header field intended for the **email software** used by the sender to identify itself (as advertising or whatever)
- Since much **junk email** is sent with mailers invented for the purpose, this field can provide much useful fodder for filters

X-Distribution:

- In response to problems with **spammers** using his software, the author of Pegasus Mail added this header
- Any message sent with Pegasus to a sufficiently large number of recipients has a header that says "**X-Distribution: bulk**". It is explicitly intended as something for recipients to filter against



X-PMFLAGS:

- This is a header added by Pegasus Mail; its **semantics** are not obvious
- It appears in any message sent with Pegasus, so it does not obviously convey any **information** to the **recipient** that is not covered by the X-Mailer: header

List of Common Headers (Cont'd)



List of Common Headers



X-Headers

- ⓘ X-headers is the generic term for headers starting with a capital X and a hyphen
- ⓘ The convention is that X-headers are **nonstandard** and provided for information only
- ⓘ This convention is frequently violated



X-Sender

- ⓘ It is the usual email analogue to the Sender: header in Usenet news.
- ⓘ This header purportedly identifies the sender with **greater reliability** than the From: header
- ⓘ In fact, it is nearly as easy to forge, and should therefore be viewed with the same **sort of suspicion** as the From: header



X-UIDL

- ⓘ This is a unique identifier used by the **POP protocol** for retrieving email from a server
- ⓘ It is normally added between the **recipient's email server** and the **recipient's actual email software**
- ⓘ If email arrives at the email server with an X-UIDL: header, it is probably junk (there is no conceivable use for such a header, but for some unknown reason many spammers add one)

Module **Flow**



**Email System
Basics**



Email Crimes



Email Headers



**Steps to
Investigate**



**Email Forensics
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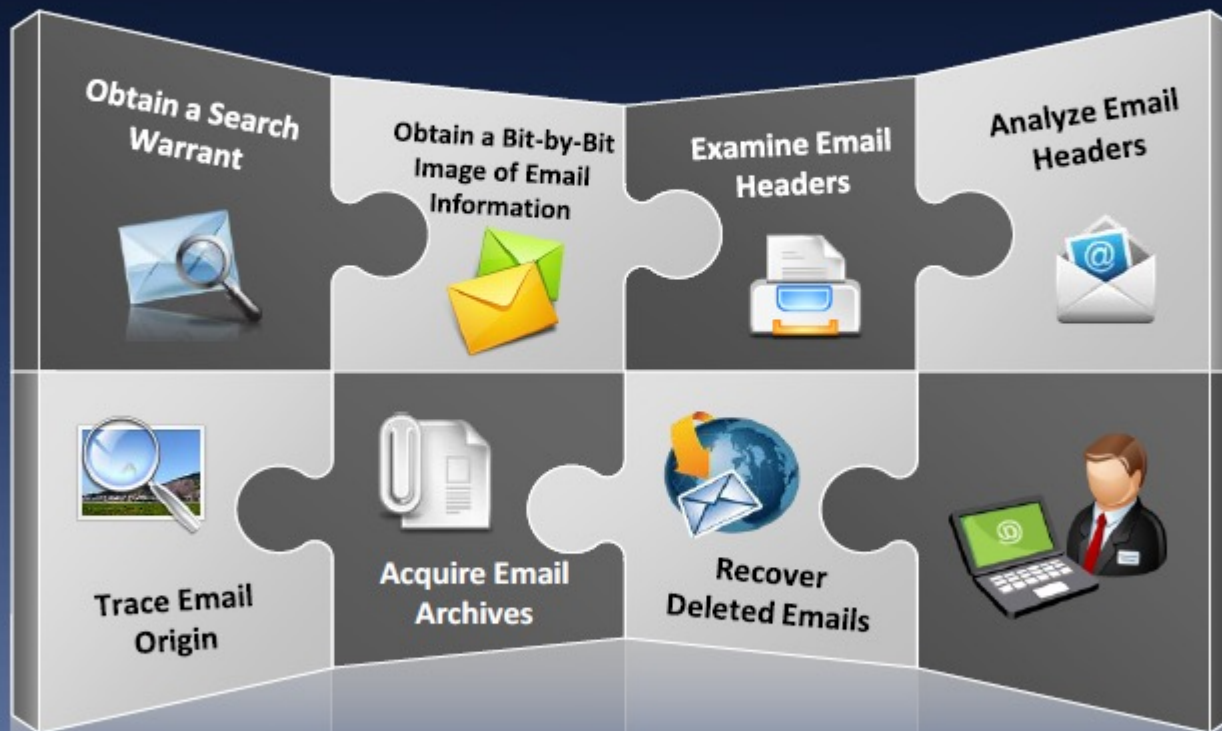
Why to Investigate Emails

- Criminals usually use emails in the following ways:



- Thus email stands as a clue to the criminal identity and may become the evidence

Investigating **Email Crime** and **Violation**





Obtain a Search Warrant and Seize the Computer and Email Account



A search warrant application should include **proper language** to perform an on-site examination of the **computer and email server**



Conduct a **forensics test** on only that equipment for which you are permitted to do so



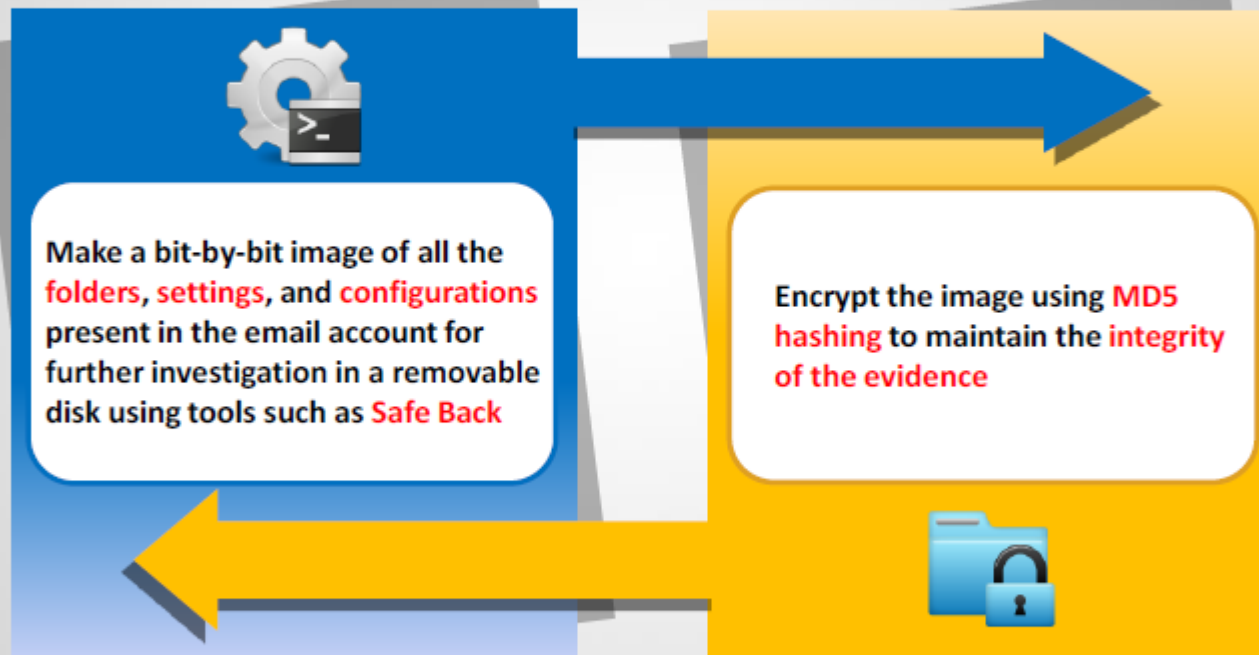
Seize the **computer and email accounts** suspected to be involved in the crime



Email accounts can be seized by changing the **existing password** of the email account either by asking the victim his/her password or from the email server



Obtain a **Bit-by-Bit Image** of Email Information



Examine **Email Headers**

Know how to find **emails headers** in various command-line, Web-based, and GUI clients

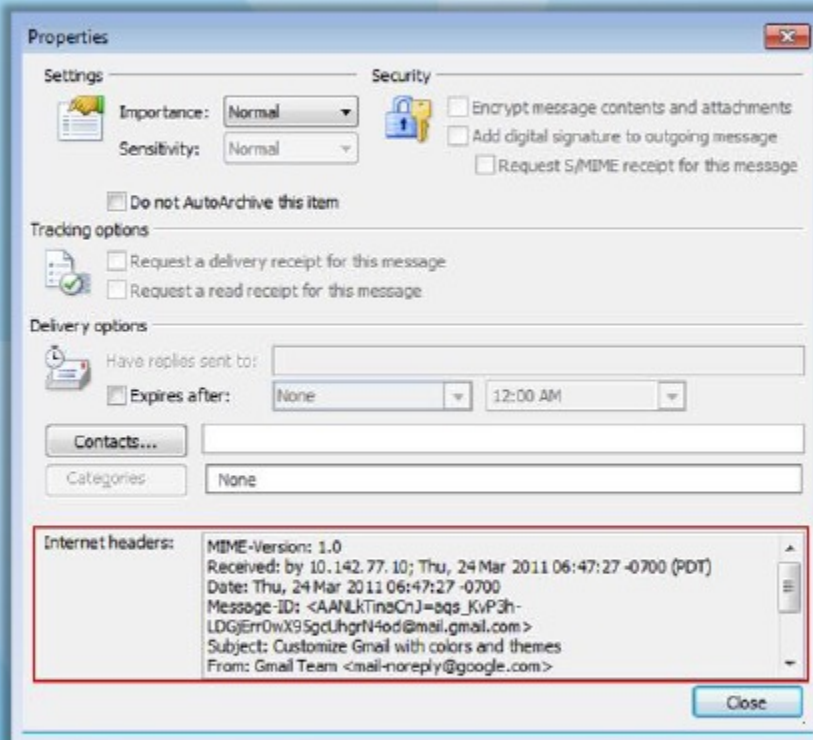


Open the email headers, **copy and paste the headers** to a text document

Headers contain significant information such as **Message sent time**, unique identifying numbers and the IP address of the sending server



Viewing Email Headers in Microsoft Outlook

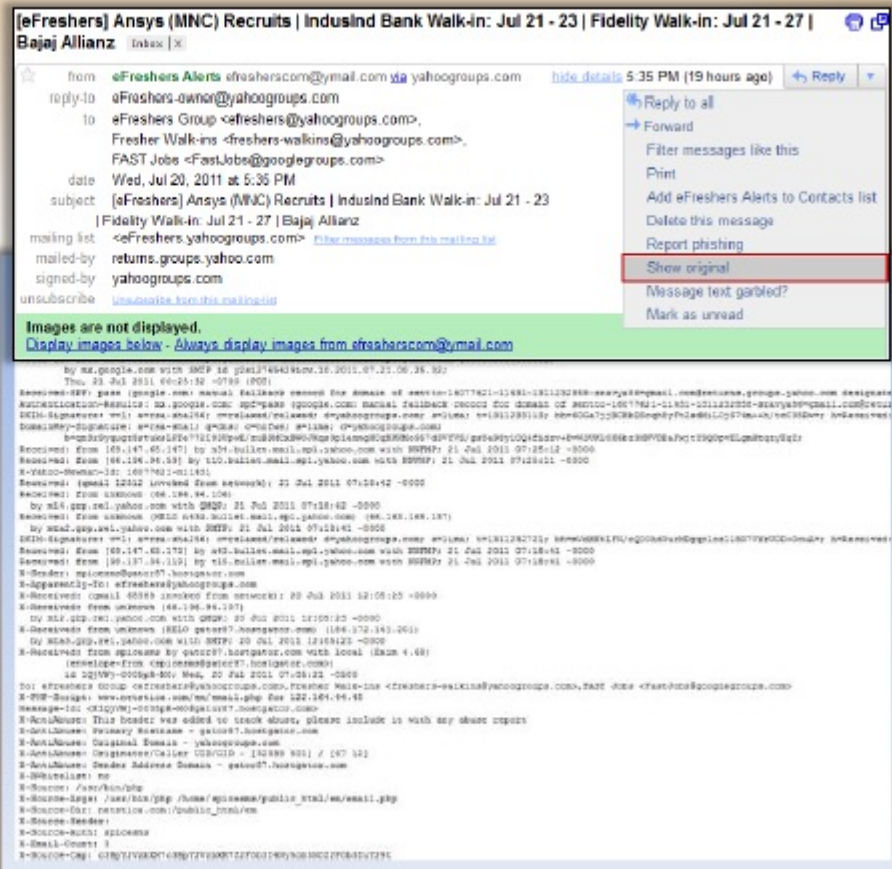


- Log on to Microsoft Outlook and open the received email
- Click on File → Info → Properties
- Select message header text, copy and paste the text in any text editor and save the file
- Sign out of the Microsoft Outlook account



Viewing Email Headers in Gmail

- Log on to **Gmail** and open the received email
- Click on the **Reply** drop-down button and navigate to the **Show original** option
- Select Message Headers - Full text and copy it
- Paste the text in any **text editor** and save the file
- Sign out of the Gmail account



Forging Headers

Another trick used by email forgers is to add **spurious** Received: headers before sending the **offending mail**

This means that someone reading the lines from top to bottom, **tracing the history of the message**, can safely throw out anything after the first forged line; even if the Received: lines after that point look plausible, they are guaranteed to be forgeries

This means that the **hypothetical email** sent from turmeric.com might have Received: lines that look something like this:

- Received: from galangal.org ([104.128.23.115]) by mail.bieberdorf.edu (8.8.5)
- Received: from nowhere by fictitious-site (8.8.3/8.7.2)...
- Received: No Information Here, Go Away!

Since the sender has no control over the message once it leaves turmeric.com, Received: headers are always added at the top and the **forged lines at the bottom of the list**

Obviously, the last two lines are complete nonsense, written by the sender and attached to the message before it was sent

Analyzing Email Headers (Cont'd)

- Gather supporting evidence as given below from the email headers and track the suspect



Return path



Recipient's
email address



Type of sending
email service



IP address of
sending server



Name of the
email server



Unique message
number



Date and time
email was sent



Attachment
files information

Analyzing Email Headers

Consider an example: Rudy sends an Email to Timmy

From: rudy@bieberdorf.edu (Rudy)

To: timmy@immense-isp.com

Date: Tue, July 05 2011 14:36:14 PST

X-Mailer: Loris v2.32

Subject: Lunch today?



Received: from mail.bieberdorf.edu (mail.bieberdorf.edu [124.211.3.78]) by mailhost.immense-isp.com (8.8.5/8.7.2) with ESMTP id LAA20869 for <timmy@immense-isp.com>; Tue, 05 July 2011 14:39:24 -0800 (PST)

Received: from alpha.bieberdorf.edu (alpha.bieberdorf.edu [124.211.3.11]) by mail.bieberdorf.edu (8.8.5) id 004A21; Tue, July 05 2011 14:36:17 -0800 (PST)
From: rudy@bieberdorf.edu (R.T. Hood)
To: timmy@immense-isp.com
Date: Tue, July 05 2011 14:36:14 PST
Message-Id: <rth031897143614-00000298@mail.bieberdorf.edu>
X-Mailer: Loris v2.32
Subject: Lunch today?



Email Header Fields

- Listed below are the most important fields to check in order to **trace the message** back to its origin and to **identify the route** it took from the **original sender**:

Field Name	Explanation
Source/Sender Header Fields	
From	Identifies email sender, usually by name and email address
Sender	Identifies actual sender of email
Reply-To	Email address to which replies should be sent
Return Path	Path (address) back to sender
Received	Except when users reside on the same server known as a message transfer agent or MTA, every email goes through at least one intermediary server as it's routed from sender to receiver. Each such intermediary appears on its own Received line
Resent-xxx	Applies to re-sent messages for From, Sender and Reply-to fields

Field Name	Explanation
Date Headers	
Date	Date and time original message was sent
Resent date	Date and time re-sent message was sent

Field Name	Explanation
Destination Header Fields	
To	Identifies name and email address for recipient
Cc	Secondary message recipients
Bcc	Blind carbon-copy message recipients
Resent-xxx	Applies to re-sent messages for To, Cc, and Bcc fields

Field Name	Explanation
Optional Headers	
Subject	Topic for message
Message-ID	Unique message identifier
In-reply-to	Identifies message being replied to
References	Identifies other messages to which this message applies
Keywords	Keywords to help sort and organize message contents
Comments	Text comments about message
Encrypted	Indicates message content is encrypted
X-xxx	Identifies user-defined fields

Received: Headers

Received: headers provide a **detailed log of a message's history**, and so make it possible to draw some conclusions about the **origin of a piece of email** even when other headers have been forged



If, for instance, the machine **turmeric.com**, whose IP address is 104.128.23.115, sends a message to mail.bieberdorf.edu, but falsely says **HELO galangal.org**, the resultant Received: line might start like this:



Received: from galangal.org ([104.128.23.115]) by mail.bieberdorf.edu (8.8.5)...



Microsoft Outlook Mail



Microsoft Outlook Mail acts like a personal information manager



The email database is normally located in the `\user account\ Local Settings\ Application Data\ Microsoft\Outlook` directory



The files stored in Outlook Mail are known as ***.pst files**



The .pst files have **archives** of all folders such as Outlook, Calendar, Drafts, Sent Items, Inbox, and Notes

Examining Additional Files (.pst or .ost Files)

1

Email messages are saved as files either on the **client computer** or **server**



2

Microsoft Outlook maintains email in **.pst** or **.ost** files



3

Online email programs such as AOL, Hotmail, and Yahoo store Email messages in folders such as **History**, **Cookies**, and **Temp**



4

Most of the email programs also include an **electronic address book**



5

Unix stores email messages as per the user



Checking the Email Validity

- Email Dossier is an online tool used to check the email validity and investigate email



Email Dossier Investigate email addresses

email address

user: 202.85.115.100 [anonymous] 47/50
log in | get account [Contact Us](#)

Validating @gmail.com...

Validation results

confidence rating: 2 - DNS
The email address passed this level of validation before the validation stopped due to the error below. This does not indicate a problem with the address, but it does not guarantee a good address, either. [more info](#)

error at level 3: UnfavorableReplyCode - SMTP server returned an unfavorable

canonical address: <@gmail.com>

MX records

preference	exchange	IP address (if included)
5	gmail-smtp-in.l.google.com	[74.125.159.27]
10	alt1.gmail-smtp-in.l.google.com	[74.125.113.27]
20	alt2.gmail-smtp-in.l.google.com	[74.125.115.27]
30	alt3.gmail-smtp-in.l.google.com	[209.85.227.27]
40	alt4.gmail-smtp-in.l.google.com	[74.125.77.27]

SMTP session

[Contacting gmail-smtp-in.l.google.com [74.125.159.27]...]
[Connected]
220 mx.google.com ESMTP 1641764881vba-T3



<http://centralops.net>

Examine the Originating IP Address



Whois.Net
DOMAIN-BASED RESEARCH SERVICES

Tool: Whois By IP Address

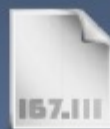
Find Out Whois By IP:
Enter IP Address: Search

[Querying whois.apnic.net]
[whois.apnic.net]
* [whois.apnic.net node-3]
* Whois data copyright terms http://www.apnic.net/db/dbcopyright.html

inetnum: 203.104.16.0 - 203.104.23.255
netname: yahoo-india
descr: Yahoo! Web Services India Pvt Ltd, Internet Content Provider
country: IN
admin-c: YI17-AP
tech-c: YI17-AP
mnt-routes: MAINT-IN-Yahoo
mnt-by: APNIC-IN
status: ASSIGNED PORTABLE

apnic: VERIFIED SCHUYDE
org-p1: APNIC-IN
org-longw: MYILL-IN-IP00
org-c1: A111-Y5
org-c2: A112-Y5
country: IN
descr: IP000 AND DELAYED IP000 FOR THE INTERNET CONTENT PROVIDER
orgname: APNIC-INDIA
orgnum: 203.104.16.0 - 203.104.23.255

<http://tools.whois.net>



Search the IP in the **whois** database



1

2

3

Collect the **IP address of the sender** from the header of the received email

Look for the **geographic address of the sender** in the whois database

Tracing Back

1

The first step in **tracing back fake mail** is to view the header's information

2

The header will show the **originating mail server**, ex: mail.example.com

3

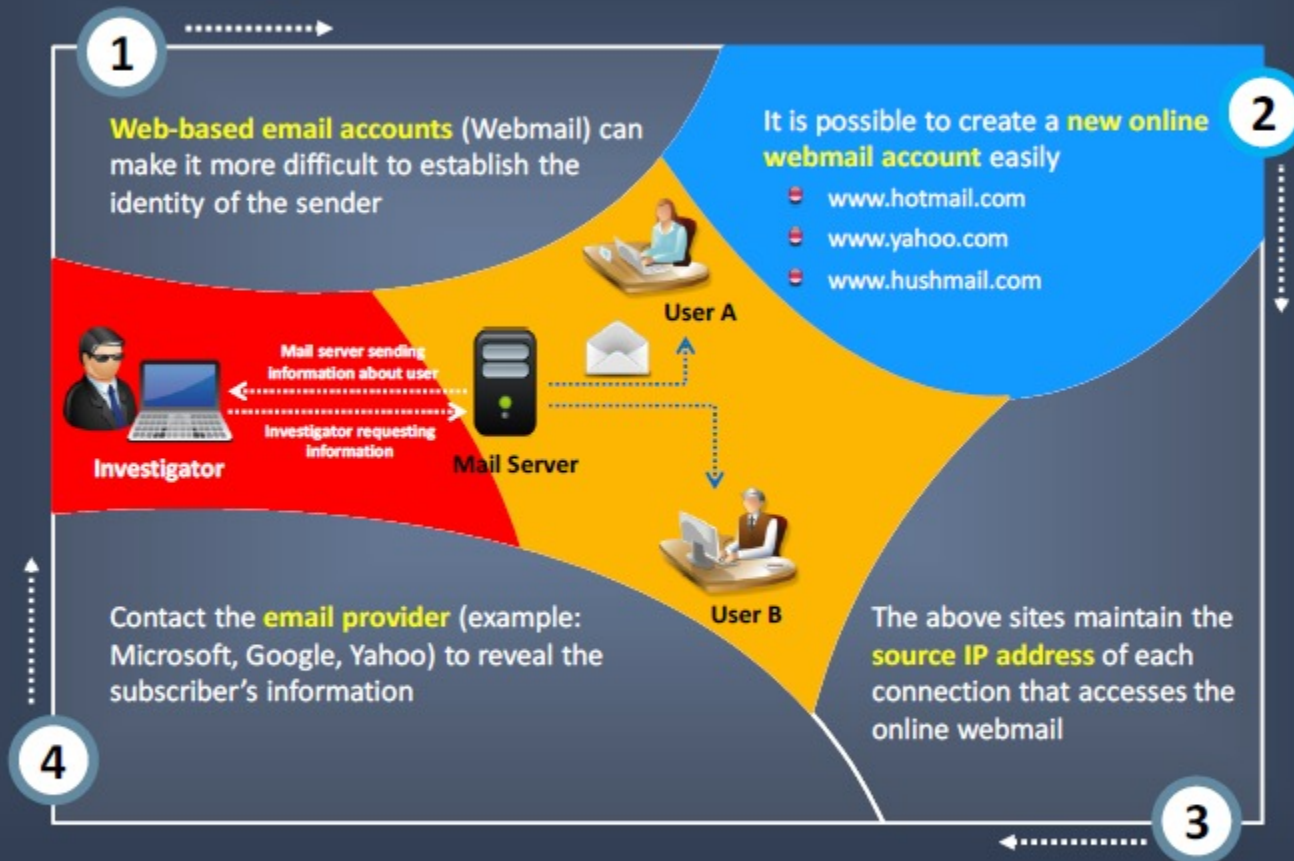
With a court order served by **law enforcement** or a **civil complaint** filed by attorneys, obtain the log files from mail.example.com to determine who sent the message

4

Information regarding the **Internet domain registration** can be found from:

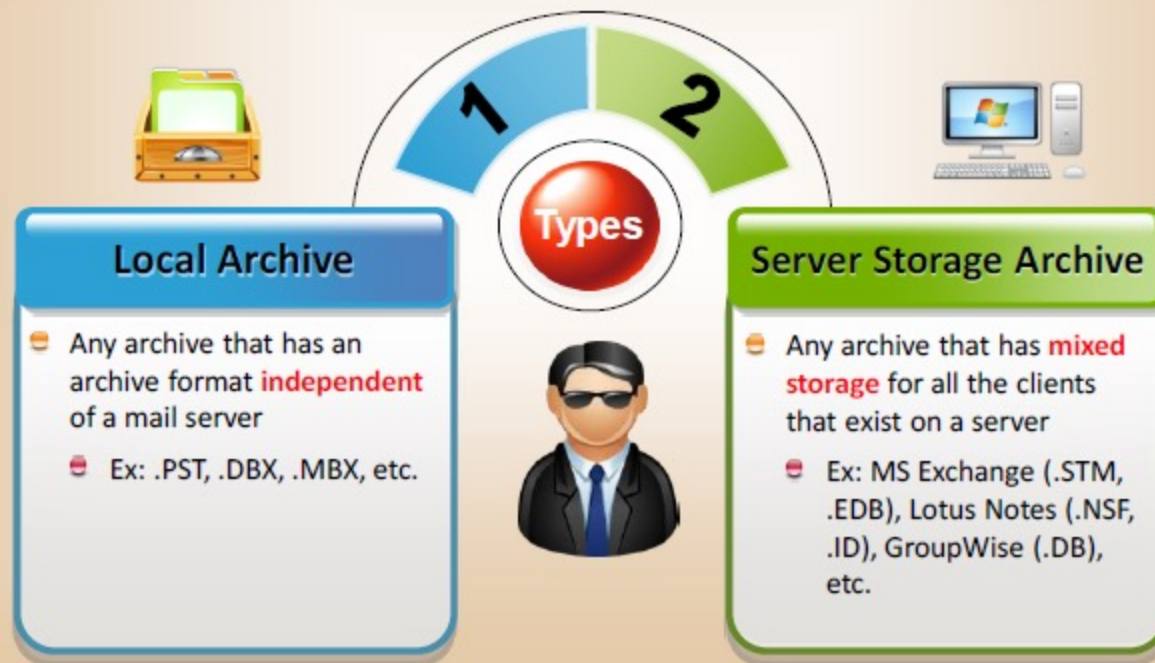
- www.arin.net
- www.internic.com
- www.freeality.com

Tracing Back Web-Based Email

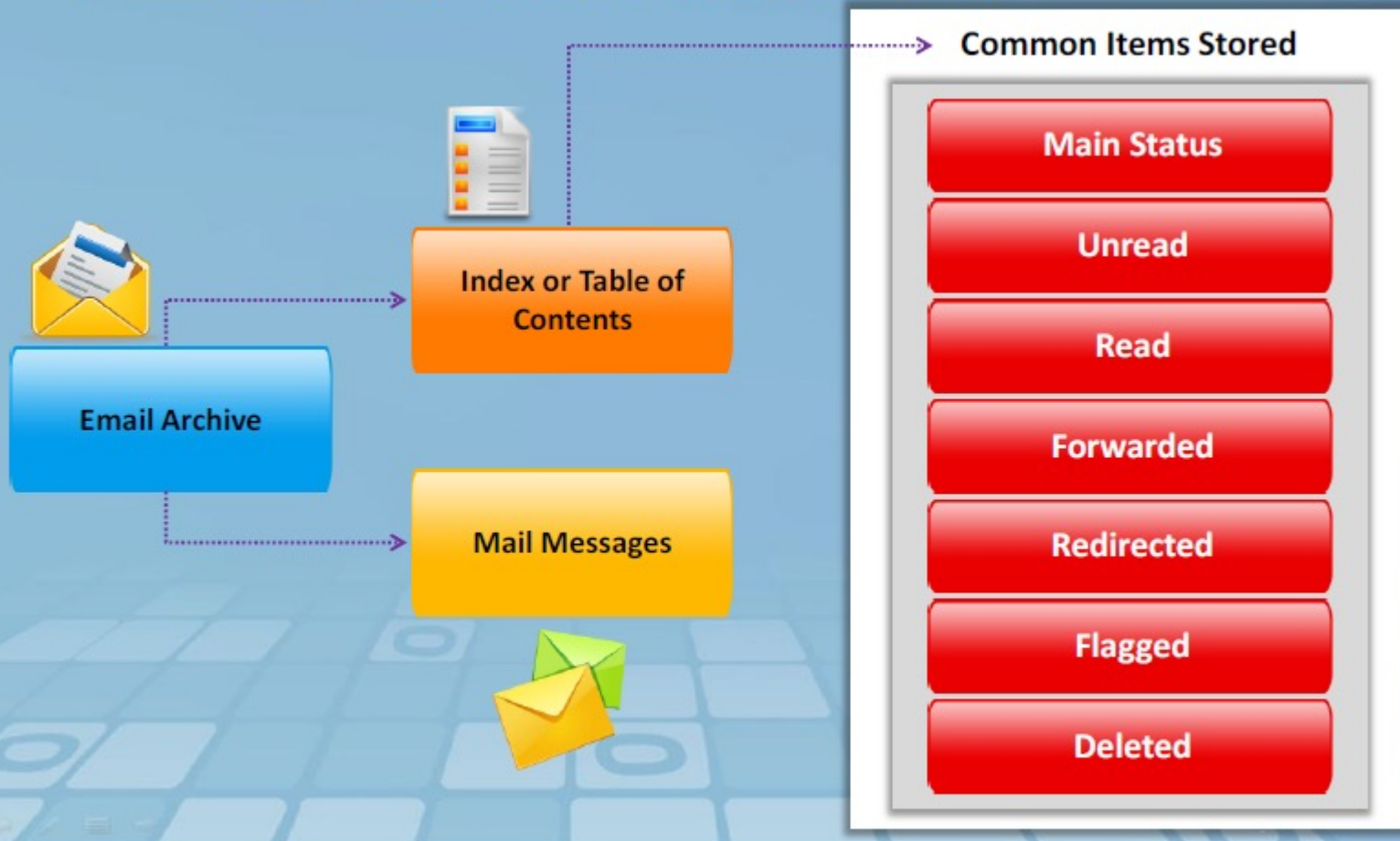


Email Archives

- Email archiving is a systematic approach to **save and protect the data contained in emails** so that it can be accessed fast at a later date
- There are two main archive types, namely:

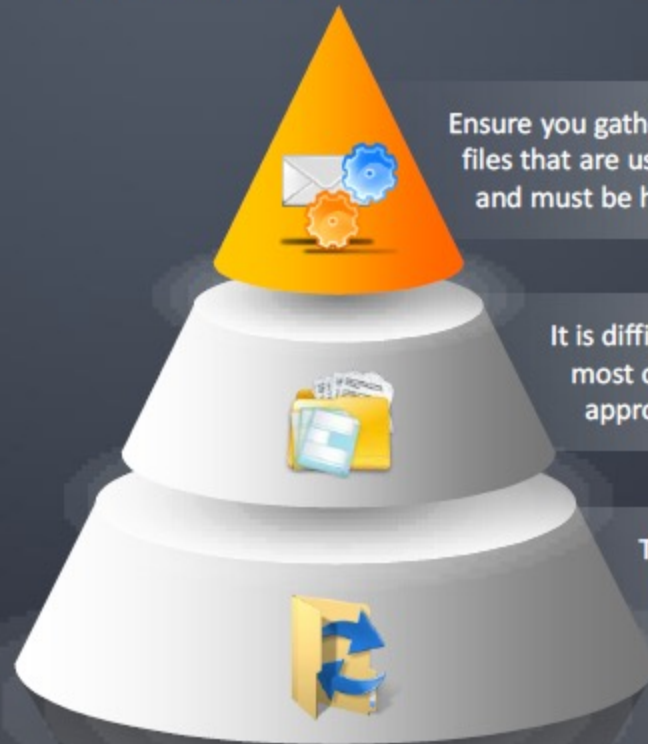


Content of Email Archives



Local Archive (Cont'd)

Local level archives are under the **control of the end user**. Follow the proper guidelines while dealing with local archives



Ensure you gather the entire archive; the **local archives** can be split into multiple files that are used to **store data**. Each of these files contain **potential evidence** and must be handled carefully

It is difficult to deal with the **webmail** as there is no **offline archive** in most cases. So consult your counsel on the case as to the best way to approach and gain access to the required data on servers

The figure on the next page shows the **types of files associated** with the **email archives** and should assist **investigators** in determining what to look into

Local Archive (Cont'd)

Common Email Local Storage Archives

The Batl

Index: *.tbi
Messages: *.tbb

The Batl ,v1.42

Index: *.tbi
Messages: *.msb

Forte Agent

Index: *.idx
Messages: *.dat

Pegasus

Index: *.pml
Messages: *.pmm

FoxMail

Index + Messages: *.dbx or
.MailDB

Outlook Express v5-6

Index + Messages: *.dbx or
MailDB

MS Outlook

Index + Messages: *.pst

Outlook Express v4.x

Index: *.idx
Messages: *.mbx

Eudora

Index: *.toc
Messages: *.mbx

Poco

Index: *.idx
Messages: *.mbx

Netscape v6.x/7.x/Mozilla

Index: *.msf
Messages: *. (No extension)

Netscape v6.x

Index: *.snm
Messages: *. (No extension)

Server Storage Archives

Microsoft Exchange

priv.edb, pub.edb, priv.stm

Lotus Notes

*.nsf, *.id

Groupwise

*.db



Example of an email archive card provided by Paraben

Local Archive

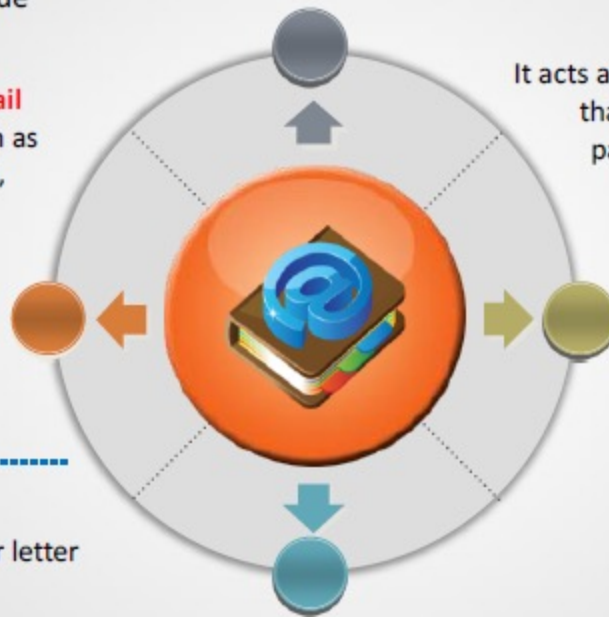
Process all items with complete structure of: header, body, encoding, attachment to compute verification through hash value

1. Header

- It is the envelope of the **email containing information** such as sender and receiver address, subject, time of creation, delivery stamps, message author, CC, and BCC
- All the above mentioned data cannot be found on all **email messages**

2. Body

- It is the **primary content** or letter of the message



3. Encoding

It acts as a **universal translator** for the email that allows different email programs to pass data to one another, though they are not the same

Types of encoding in emails:

MIME (Multipurpose Internet Mail Extensions)

It is a protocol that allows **non-ASCII files** (video, graphics, and audio) to be built in the email message

UUCODE

UNIX format for attachment encoding

BINHEX

Mac format for attachment encoding

4. Attachment

It is an extra item that comes as a supplement to the body

Server Storage Archive (Cont'd)

Lotus Notes

Follow the guidelines when dealing with Lotus Notes

- Gather the ***.NSF** file
- Gather the **associated *.ID** file for the archive, this is the encryption key that allows you to **open encrypted mail** that is stored



Novel GroupWise

Follow the guidelines when dealing with Novel GroupWise

- Ensure that the **acquisition** is done on the entire directory and the **structure remains intact**
- **Ngwguard.db** is stored in the root of the email directory and is the key file of the GroupWise structure. It tells the GroupWise about each user account and where it is located
- Other key files include **wphost.db** and **gwcheck.db**, but the entire directory must be intact to do an examination

Server Storage Archive

MS Exchange

Follow the guidelines when dealing with MS Exchange

- Do not deal with an **active Exchange server**. Take a backup of the server, this maintains the best date structure for the data
- Gather all the **data files** associated with the server. PRIV.EDB, PUB.EDB, and PRIV.STM files create the complete archive

It is a **rich text database file** containing message headers, message text, and standard attachments



PRIV.EDB

It is a database file to store public folder **hierarchies and contents**



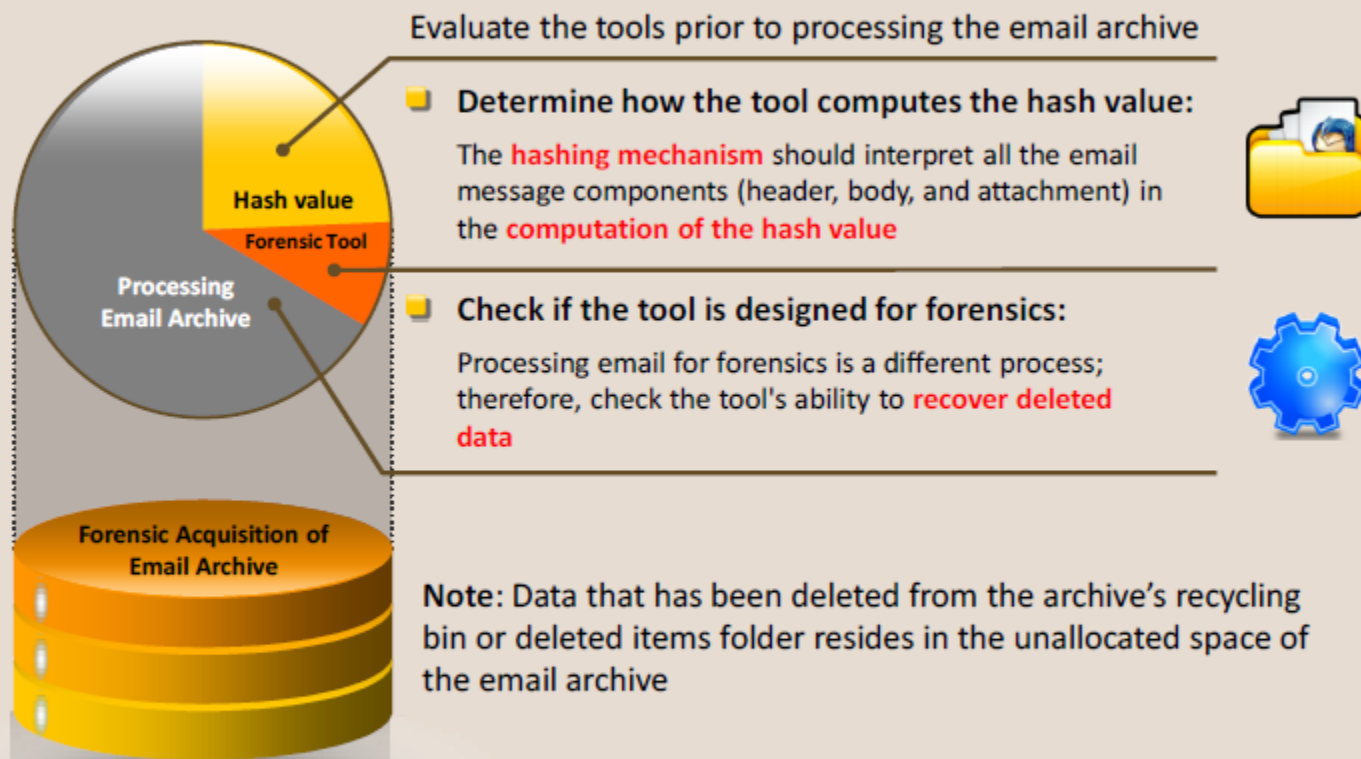
PUB.EDB

It is a streaming Internet content file containing video, audio, and other media that are formatted as streams of **Multipurpose Internet Mail Extensions**



PRIV.STM

Forensic Acquisition of Email Archive (Cont'd)



Forensic Acquisition of Email Archive (Cont'd)

Processing Local Email Archives:

 Outlook PST files and Outlook Express are the most common email archives and can be found on the desktop system

1. Outlook PST File Acquisition:

- Acquire a bit-stream image of the entire drive and then extract the PST file from the **drive image using multiple tools**
- Once the file is extracted, choose tools such as **Paraben's Email Examiner** to process the proprietary email archive into usable messages

2. Outlook Express Acquisition:

- Outlook Express stores message information in a **DBX** format
 - Install Outlook Express on the **analysis machine**
 - Find the Outlook Express DBX files on the suspect image
 - Copy all **DBX files** from the suspect machine's directory to the analysis machine's Outlook Express directory
 - Disconnect the analysis computer from the network to prevent **accessing mailboxes** involuntarily
 - Open Outlook Express on the analysis machine
- Dedicated viewer such as **OE Viewer** can also be used to view the DBX files

Forensic Acquisition of Email Archive

Processing Server Level Archives:



When processing a **server level archive**, there are many files to look into. Gather different data based on the email server used



Acquisition stage for a server archive is different to the **local archive acquisition**. Here you acquire the appropriate files where the archive is stored



Wide range of tools are not available for **network level archives**. However there are other tool options that are designed to restore archives for review



Ontrack PowerControls assist administrators to copy, search, recover, and analyze email and other mailbox items directly from Microsoft Exchange Server backups, un-mounted databases (EDB) and information store files



Another tool is **Paraben's Network Email examiner** that processes MS Exchange archives as well as GroupWise and LotusNotes

Deleted Email Recovery

- Recovery of deleted email messages varies based on the email client used. Examples of how deleted email works:



Eudora Mail

- Messages for deletion are **tagged for deletion** and no longer visible in the mailbox
- However, messages reside in the **trash folder** until the trash folder is emptied



Outlook PST

- Data is taken from the **active part of the archive** to a recycle bin
- If the recycle bin is emptied, it will go to the **unallocated space** of the email archive where it resides for a period of weeks
- Recovery of this data varies depending on the **size of the archive**

Module **Flow**



**Email System
Basics**



Email Crimes



Email Headers



**Steps to
Investigate**



**Email Forensics
Tools**



**Laws and Acts
against Email Crimes**

Stellar Phoenix Deleted Email Recovery

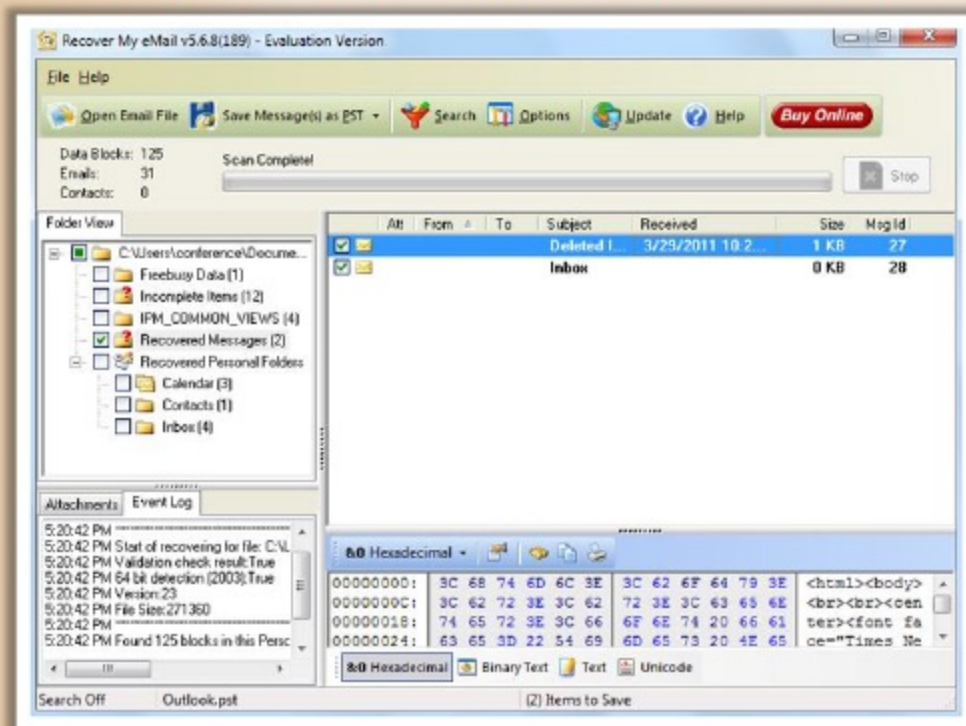
- Stellar Phoenix Deleted Email Recovery is the tool to **recover permanently removed/deleted PST items and DBX emails**
- The deleted email recovery software restores PST items as **pst**, **.eml**, and **.msg** files and DBX emails as **.dbx** and **.eml** files



<http://www.stellarinfo.com>

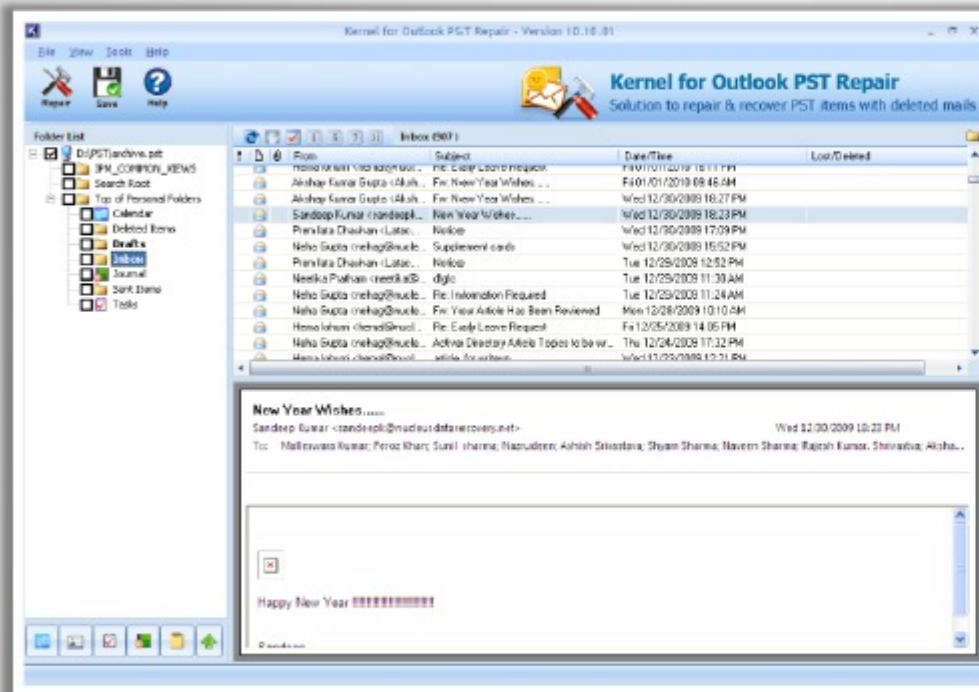
Recover My Email

- Recover My Email is email recovery software that can **recover deleted email** messages from either Microsoft **Outlook PST** file or Microsoft **Outlook Express DBX** files



<http://www.recovermyemail.com>

Outlook Express Recovery



Outlook Express Recovery tool **recovers emails** and other items from damaged or corrupt Outlook Express 5.0 and above **.DBX Files**

Features

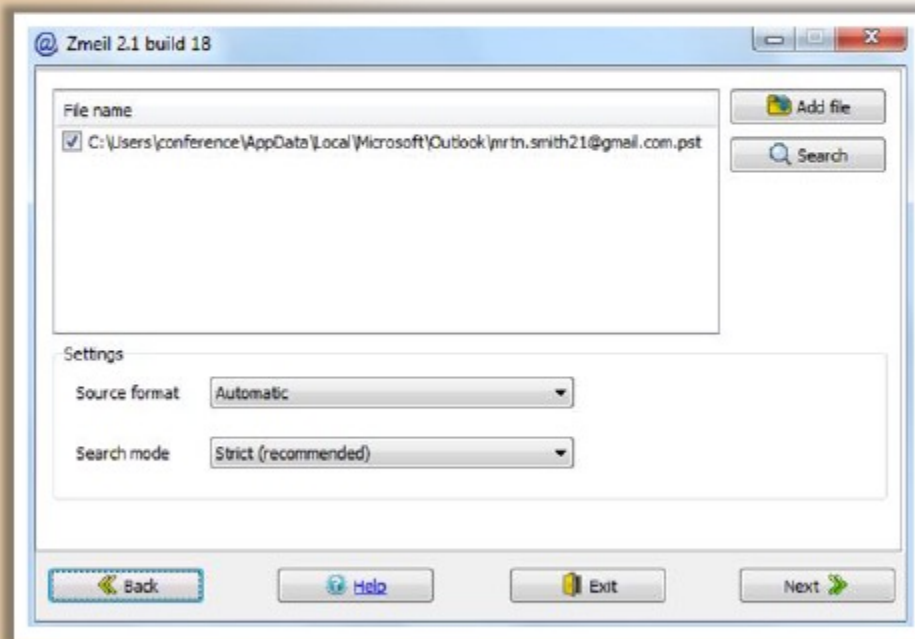
- Recovers **permanently deleted** Outlook Items
- Recovers emails from **corrupt PST** file
- Fixes damaged PST headers** and recovers all types of items from the PST file



<http://www.recoveryemail.com>

Zmeil

- Zmeil **recovers deleted email messages** or messages from corrupt email databases in case of accidental deletion or database corruption, caused by software problems or power spikes



<http://www.z-a-recovery.com>



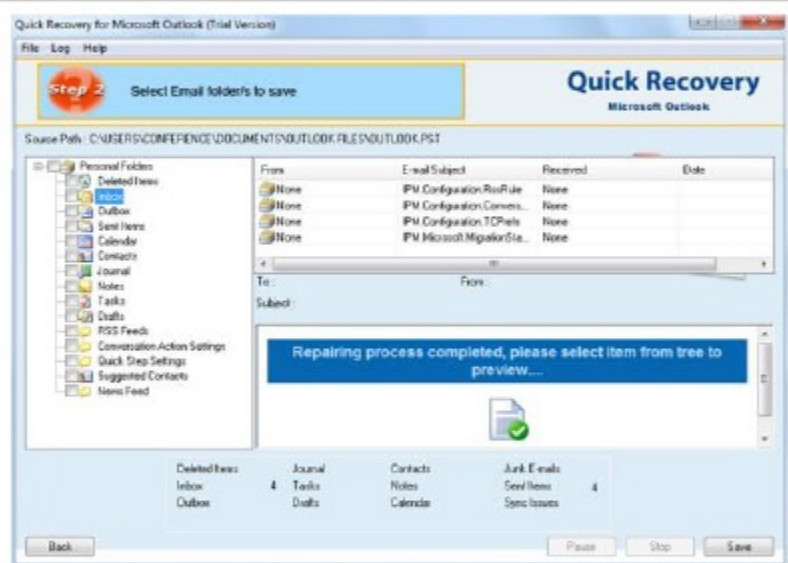


Quick Recovery for MS Outlook

Quick Recovery for MS Outlook repairs and restores data from **damaged or corrupted Microsoft Outlook files**

It retrieves emails from corrupt or damaged PST files

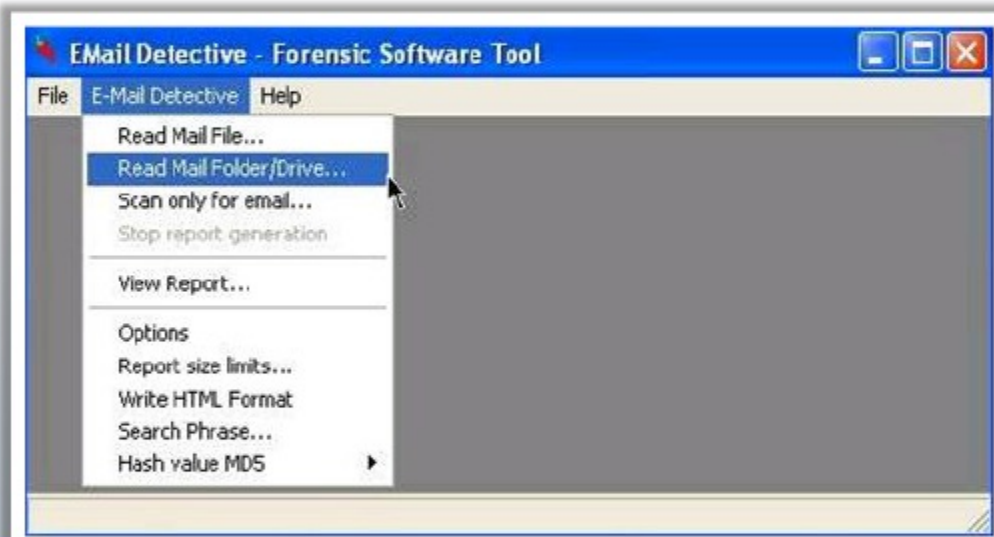
It repairs corrupt PST Files and recovers all Microsoft Outlook Items such as Calendar, Tasks, Notes, Contacts, Journals etc.



<http://www.recoveryourdata.com>

Email Detective

Forensic software tool that is used in several investigations and data recovery



Email Trace - Email Tracking

- Email trace tool helps to **track the email sender** and IP address of the sender
- To trace an email:
 - Open the received email and copy the headers
 - Go to http://www.ip-adress.com/trace_email/
 - Paste the email message headers
 - Click on **"Trace Email Sender"**
 - Email sender IP address location and IP address information are traced



To Trace An Email, Copy And Paste The Email Header Below:

```
Delivered-To: prashant@ircotc.co.in  
Received: by 10.68.40.102 with SMTP id w6cs50843pbkz  
    Fri, 29 Jul 2011 23:54:13 -0700 (PDT)  
Received: by 10.68.36.135 with SMTP id q7ar1092346pbj.222.131200852622;  
    Fri, 29 Jul 2011 23:54:12 -0700 (PDT)  
Return-Path: <ticketadmin@ircotc.co.in>
```

Trace Email Sender

Clear



Email Trace - Email Tracking



Email Tracing successful!

At Sat, 30 Jul 2011 12:24:11 +0530 (IST), someone sent you an email from the IP address 202.93.154.75 located in India, Delhi.

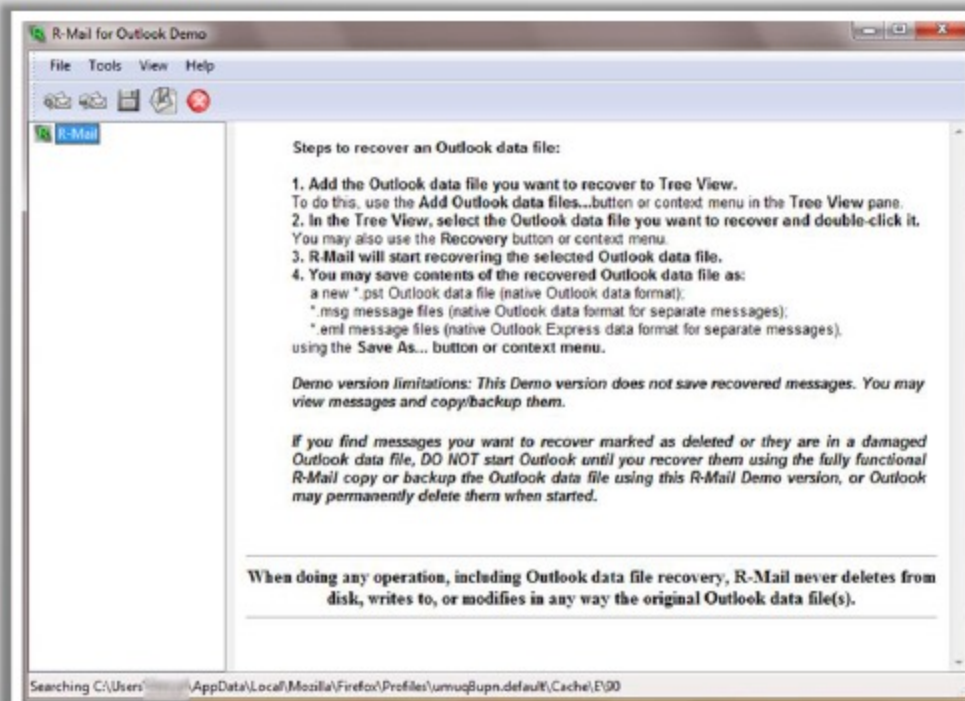
IP address:	202.93.154.75
IP address country:	India
IP address state:	Delhi
IP address city:	Delhi
IP address latitude:	28.6967
IP address longitude:	77.2167
ISP of this IP:	IRCTC
Organization:	IRCTC
Local Time of this IP country:	2011-07-30 15:25



<http://www.ip-adress.com>

R-Mail

- R-Mail for Outlook **recovers accidentally deleted Outlook email messages**, contacts, notes, tasks, and other items and **repairs damaged Outlook data files (*.pst)** where Outlook stores folders with the data



<http://www.r-tt.com>

FINALEMAIL

FINALEMAIL recovers the **email database files** and locates lost emails that do not have **data location information** associated with them



Recovers **email messages** and **attachments** emptied from the Deleted Items folder in Microsoft Outlook Express, Netscape Mail, Eudora



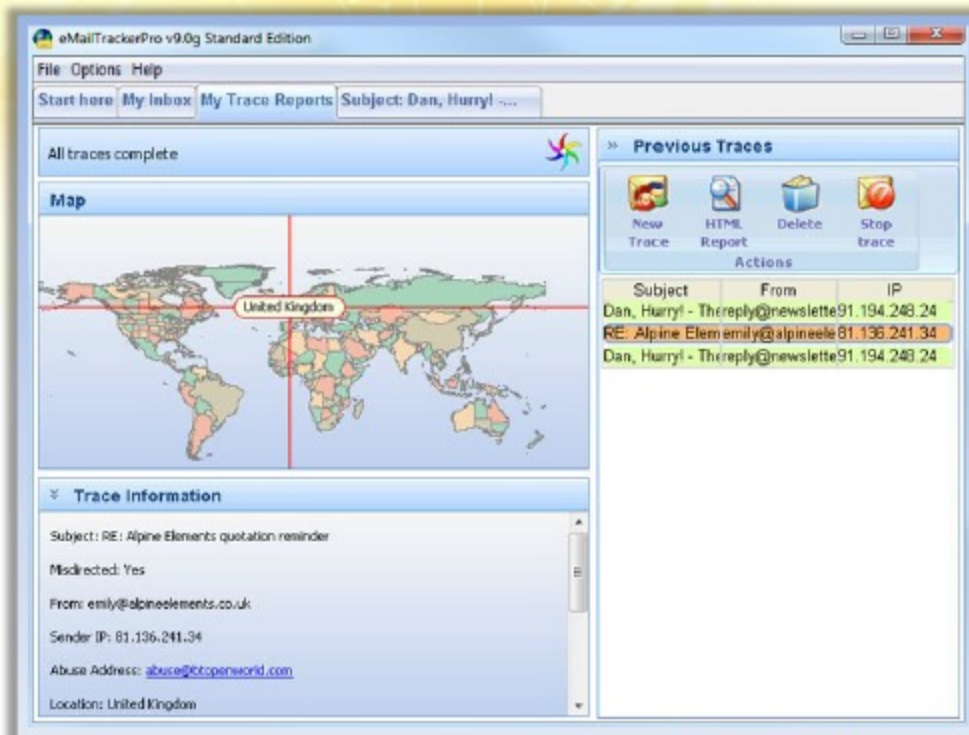
It can recover **entire email database files**



<http://finaldata2.com>

eMailTrackerPro

- eMailTrackerPro analyzes the **email header** and provides the **IP address of the machine** that sent the email



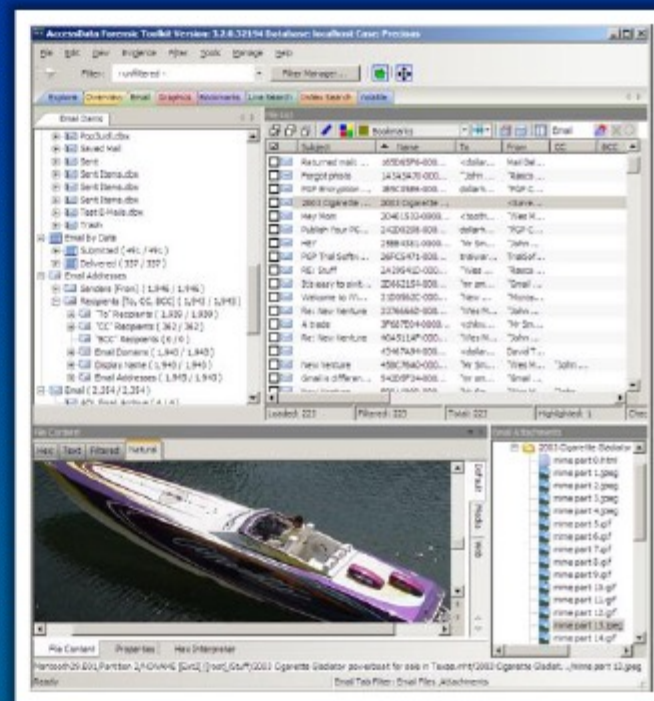
<http://www.emailtrackerpro.com>

Forensic Tool Kit (FTK)

- AccessData FTK is a forensic tool known to perform email analysis

- **Features**

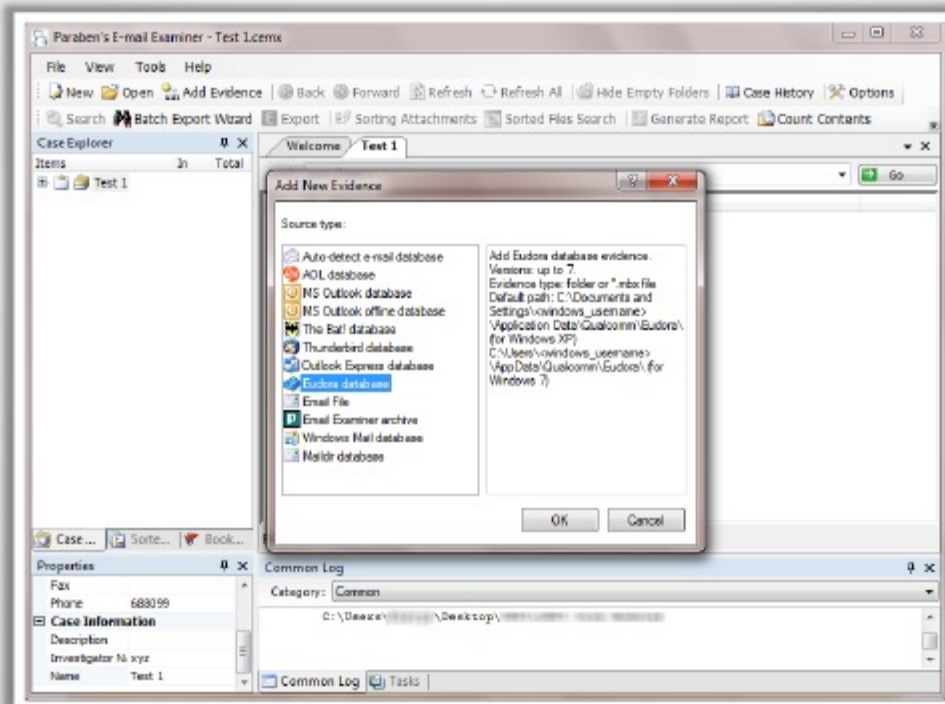
- **Email analysis supports** Outlook, Outlook Express, AOL, Netscape, Yahoo, Earthlink, Eudora, Hotmail, and MSN email
- **View, search, print, and export email messages and attachments**
- **Recovers deleted and partially deleted emails**



<http://accessdata.com>

Paraben's E-mail Examiner

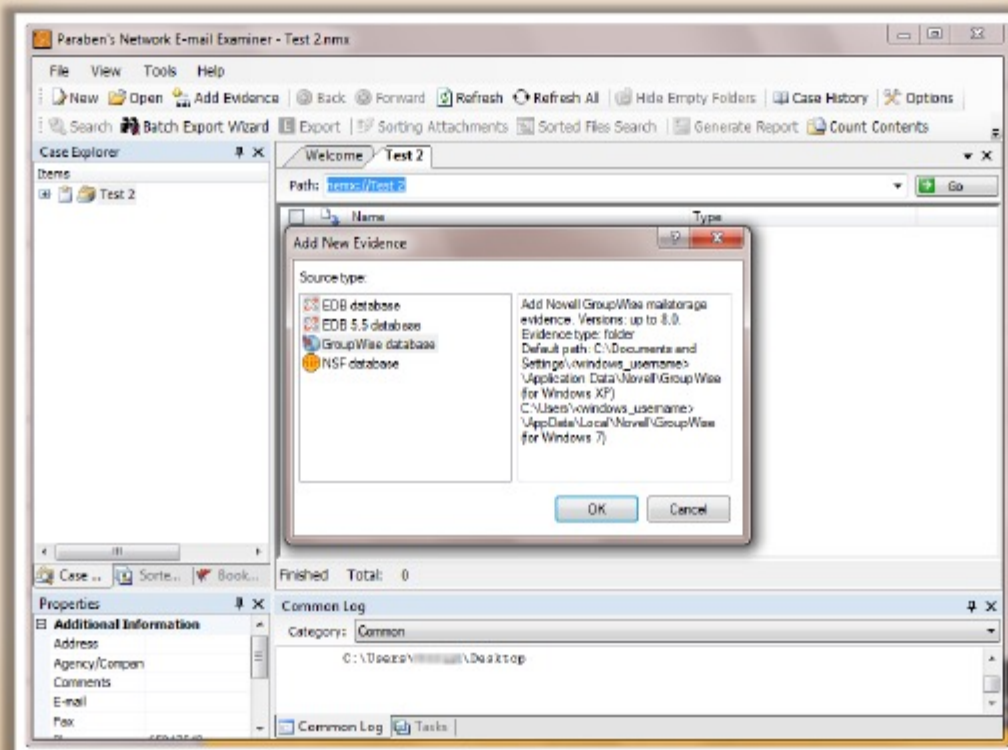
- It forensically examines email formats such as America Online (AOL), Outlook Exchange (PST), Eudora, etc.
- It recovers deleted messages and folders



<http://www.paraben.com>

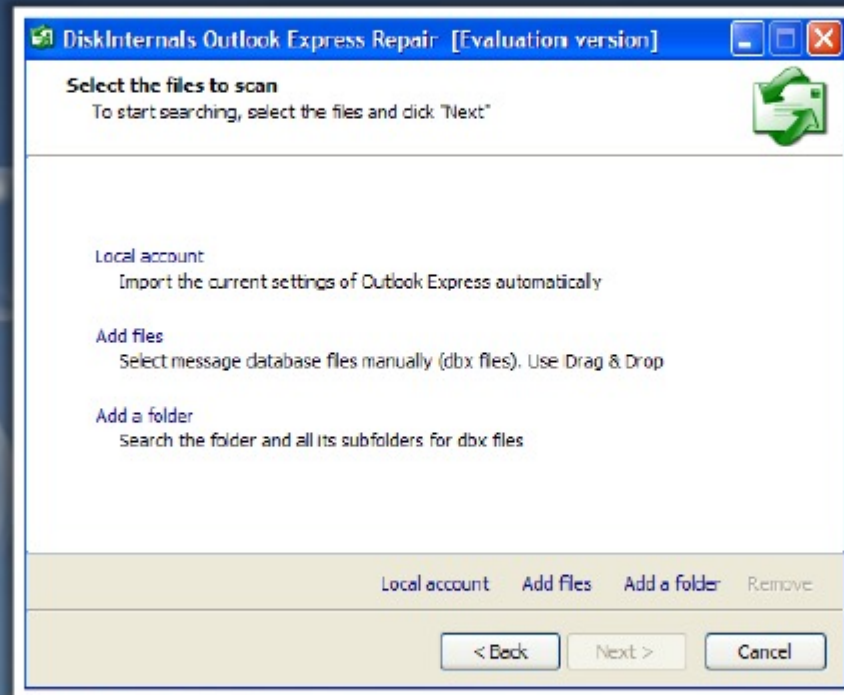
Paraben's Network E-mail Examiner

It helps examine **Microsoft Exchange (EDB)**, **Lotus Notes (NSF)**, and **GroupWise** email stores



DiskInternal's Outlook Express Repair

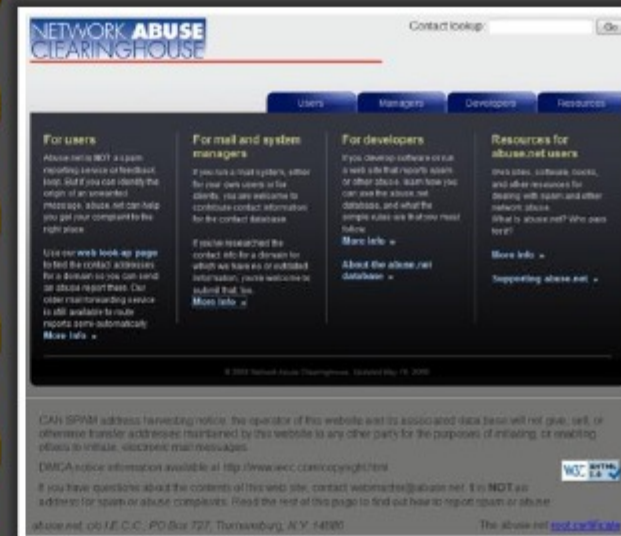
DiskInternal's Outlook Express Repair detects email accounts, scans them for **damages** and **restores** their contents as EML files



<http://www.diskinternals.com>

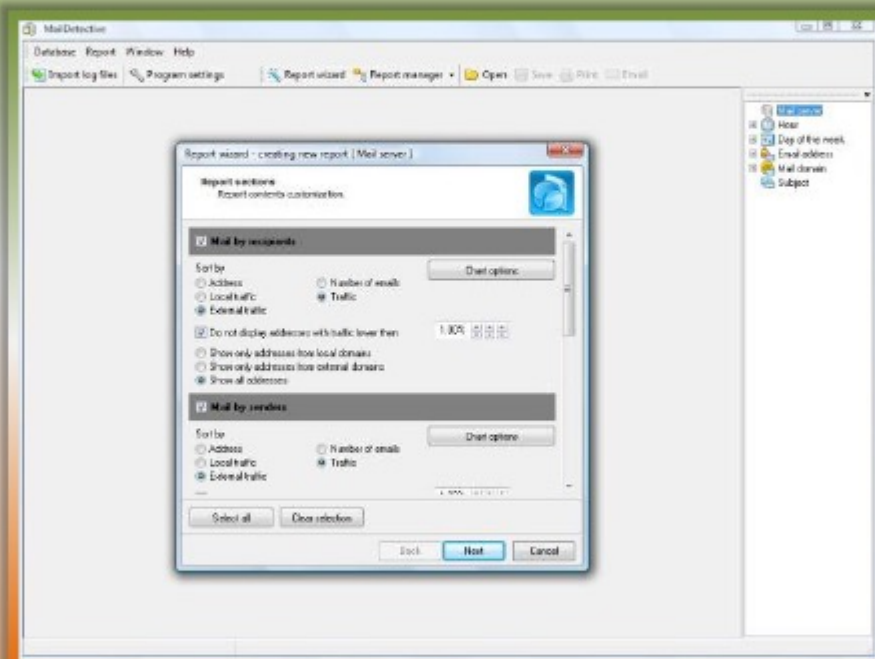
Abuse.Net

- Abuse.net helps the **Internet community** to report and control network abuse and abusive users
- It does not include **blacklist or spam analysis services**
- Once registered, when you send a message to **domain-name@abuse.net**, where **domain-name** is the name of the domain that was the source of junk email or another abusive practice, the system **automatically emails** your message to the best reporting address(es) known for that domain



MailDetective Tool

- It is a **monitoring application** designed to control email use in a corporate network
- It **analyses email server log files** and **provides detailed reports** about private and business emails coming to and from the corporate network, as well as traffic distribution by users and email addresses



<http://www.advsoft.info>

Module Flow



Email System Basics



Email Crimes



Email Headers



Steps to Investigate



Email Forensics Tools



Laws and Acts against Email Crimes

U.S. Laws Against Email Crime: **CAN-SPAM Act** (Cont'd)

- The CAN-SPAM Act of 2003 (**Controlling the Assault of Non-Solicited Pornography and Marketing Act**) establishes requirements for those who send commercial email, spells out penalties for spammers and companies whose products are advertised in spam if they violate the law, and gives consumers the right to ask emailers to **stop spamming** them

Main provisions:

It bans false or misleading header information



NO SPAM!

It prohibits deceptive subject lines

It requires that the email give recipients an opt-out method



It requires that commercial email be identified as an advertisement and include the sender's valid physical postal address

U.S. Laws Against Email Crime: **CAN-SPAM Act**

Penalties :

- Each violation of the provisions on the previous page is **subject to fines of up to \$11,000**
- Additional fines are provided for commercial mailers who not only violate the rules described on the previous page, but also:
 - "Harvest" email addresses from websites or Web services that have published a notice **prohibiting the transfer of email addresses** for the purpose of sending email
 - Generate email addresses using a "**dictionary attack**" – combining names, letters, or numbers into multiple permutations
 - Use scripts or other automated ways to register for **multiple email or user accounts** to send commercial email
 - Relay emails through a **computer or network without permission** – for example, by taking advantage of open relays or open proxies without authorization
- The law allows the DOJ to seek criminal penalties, including imprisonment, for commercial emailers who do – or conspire to:
 - **Use another computer without authorization** and send commercial email from or through it
 - Use a computer to **relay or retransmit multiple commercial email messages** to deceive or mislead recipients or an Internet access service about the origin of the message
 - **Falsify header information** in multiple email messages and initiate the transmission of such messages



18 U.S.C. § 2252A

- This law states that any person who commits any of the following crimes shall be punished as defined under this title, and imprisoned not less than 5 years and not more than 20 years.":



Knowingly **distributes, offers, sends, or provides** to a minor any visual depiction, including any photograph, film, video, picture, or computer generated image or picture, whether made or produced by electronic, mechanical, or other means

Knowingly mails, or transports or ships in **interstate or foreign commerce** by any means, including by computer, any child pornography

Knowingly **reproduces any child pornography** for distribution through the mail, or in interstate or foreign commerce by any means, including by computer

Knowingly **receives or distributes any child pornography** that has been mailed, or shipped or transported in interstate or foreign commerce by any means, including by computer



18 U.S.C. § 2252B

This law states that:



Whoever knowingly uses a **misleading domain name** on the Internet with the intent to deceive a person into **viewing material constituting obscenity** shall be fined under this title or imprisoned not more than 2 years, or both



Whoever knowingly uses a misleading domain name on the Internet with the **intent to deceive a minor** into viewing material that is **harmful to minors** on the Internet, shall be fined under this title or imprisoned not more than 4 years, or both



For the purposes of this section, a domain name that includes a word or words to indicate the **sexual content of the site**, such as "sex" or "porn", is not misleading



Email Crime Law in Washington: **RCW 19.190.020**



This law is for residents of Washington; it states that:



- No person may initiate the transmission, conspire with another to initiate the transmission, or assist the transmission, of a **commercial electronic mail message** from a computer located in Washington or to an electronic mail address that the sender knows, or has reason to know, is held by a **Washington resident** that:
 - Uses a third party's Internet domain name without **permission of the third party**, or otherwise misrepresents or obscures any information in identifying the point of origin or the transmission path of a commercial electronic mail message; or
 - Contains **false or misleading information** in the subject line

Module Summary



- ☐ Spammers obtain email addresses by harvesting addresses from Usenet postings, DNS listings, or web pages
- ☐ Chat rooms can also be used as a social engineering tool to collect information for committing several other crimes
- ☐ Phishers incite the targeted users to provide personal information in illegitimate websites
- ☐ Online email program such as AOL, Hotmail, and Yahoo store Email messages in folders such as History, Cookies, and Temp
- ☐ Headers contain significant information such as Message sent time, unique identifying numbers and IP address of the sending server

Copyright 2009 by Randy Glasbergen
www.glasbergen.com



"We can't think about bankruptcy until we've explored all of our options. Do you still have that e-mail from the Nigerian prince?"

Copyright 2003 by Randy Glasbergen.
www.glasbergen.com



**“Why do they call it E-mail? If it’s nothing
but spam, why don’t they call it S-mail?”**